Online Appendix

Taxing Multinational Digital Platforms: Evidence from France's Digital Services Tax

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A Firm identification and website linkage

This appendix documents the construction of the research roster of 35 corporate groups that are likely liable for the French DST and the linkage of those groups to websites in the WhoTracksMe (WTM) traffic universe. The work proceeds in two stages. The first stage starts from the U.S. Trade Representative's list of 27 groups, assembles a broad universe of candidate and peer firms with financial data, and uses financial screens and a business-model review to select eight additional groups. The second stage assigns each WTM domain to its December 2018 owner, groups domains into website families, and applies coverage and activity screens to obtain the liable website families used in the traffic analysis. Table A.5 at the end of the appendix summarizes the populations that result.

A.1 Starting point: the USTR list

The starting point is the table of company groups "expected to be covered" by the French DST in the USTR's Section 301 report (United States Trade Representative, 2019). The USTR drew that table from an economic-impact study commissioned from Deloitte/Taj and from the French National Assembly's report on the bill. It names 27 groups: Airbnb, Alibaba, Alphabet, Amadeus, Amazon, Apple, Axel Springer, Booking Holdings, Criteo, eBay, Expedia, Facebook, Groupon, Match Group, Microsoft, Rakuten, Randstad, Recruit, Sabre, Schibsted, Snap, Travelport, Twitter, Uber, Verizon, Wish (ContextLogic), and Zalando. Because the USTR compiled the list to support its finding that the tax discriminates against U.S. firms, it serves as a starting point that requires validation in two directions: groups on the list that may not meet the statutory criteria, and groups that meet them but are absent from the list.

A.2 Candidate universe and source databases

Candidate list. A deliberately over-inclusive screening list of 130 groups was first compiled from public sources: the USTR and Deloitte materials, the statutory text, parliamentary materials, and ministerial statements, with assistance from a large language model. The list contains all 27 USTR groups. The remaining rows are candidates to be confirmed or rejected against accounting data. Each row records the group's in-scope service category (digital intermediation, or targeted advertising and the sale of user data), a confidence tier (25 high, 39 medium, 66 low), and a threshold band. Forty-three rows are expected to clear the statutory thresholds of €750 million worldwide and €25 million in France. The other 87 were added under a lowered safety band of €500 million and €5 million. The list also retains groups that were in scope on 1 January 2019 but were later acquired, renamed, merged, or closed, and it records the legal entity that filed for the relevant year. Over-inclusion is intentional. A false positive at this stage only adds records that the screens later remove, whereas a group omitted at this stage cannot be recovered.

Industry-peer universe. Each candidate is resolved to a database identifier: listed firms by ticker to a Compustat `gvkey`, and private firms by name and country to an Orbis BvD identifier. The universe then ex-

pands to every firm with roughly €500 million or more in global operating revenue whose primary industry code falls in the union of the NAICS and SIC codes carried by the resolved candidates. Firms come from Compustat, which covers listed firms worldwide, and from Orbis, which adds private firms and firms with a French presence (S&P Global Market Intelligence, 2025; Bureau van Dijk, a Moody's Analytics company, 2025). Firms whose business model appears out of scope, such as first-party retailers, pure software providers, and subscription streaming services, are tagged rather than dropped, so that the screens can separate them later. The resulting financial panel covers 3,895 firm identifiers over 2014–2025: 105 resolved primary candidates, 3,788 industry peers, and two duplicate French-shell identifiers. Compustat amounts are converted to euros using Orbis's own exchange rates, and all screens use fiscal year 2018.

French footprint. Statutory French revenue is not reported in consolidated accounts, so each group's 2018 French footprint is reconstructed from the Orbis ownership tree as of 2018 and the unconsolidated accounts of its French entities. The default aggregation sums operating leaves of the tree and excludes holding companies where possible; a French group that reports its own activity can enter through its own accounts. Four alternatives retain operating parents, sum all directly attributed French entities, select the largest French entity, or follow the global-ultimate-owner path. These are alternative accounting footprints used to assess attribution and double counting. None of them measures turnover apportioned by the statutory user-location rules.

A.3 What the financial screens establish

The screens combine company size, industry classification, business-segment composition, and the French footprints. The screens use two industry-code sets. The broad set (49 NAICS and 24 SIC codes) is the union of the candidates' codes. The strict set (12 NAICS codes) keeps only codes that correspond to digital intermediation or advertising services. It drops codes that the broad set inherits from conglomerate candidates, such as Apple's equipment-manufacturing code or Amazon's e-commerce retail codes. The broad screen requires worldwide revenue above €750 million in a broad in-scope industry. The French screens additionally require recorded French revenue above €25 million under one of the footprint aggregations. The segment-based screen applies both thresholds to revenue scaled by the share of segment sales in strict in-scope industries. Because the broad code set covers about 97 percent of the pre-curated universe, the global size threshold, rather than the industry criterion, is what binds in the broad screen.

Table A.1 reports the screens' overlap with the USTR list. Missing segment data, unresolved French entities, and recorded French sales below €25 million account for some gaps. Agreement with the USTR list describes overlap between classification methods, rather than the sensitivity or specificity of a verified liability test.

Several cases show why documentary review remains necessary. Airbnb, Alibaba, and eBay have recorded French sales below €25 million in the report despite appearing on the USTR list. Their local accounts need not capture the revenue attributable to French users. Rakuten's French entities appear in the

Table A.1: Financial-screen overlap with the 27 USTR groups

Screen in the archived FY2018 report	Groups flagged	USTR groups recovered
Broad size and industry screen	893	25
French operating-leaf screen	118	15
French operating-parent screen	129	16
All directly attributed French entities	131	16
Largest French entity	127	16
Segment-based, operating-parent screen	50	10

Notes: Source: the archived FY2018 liability-screen report underlying the institutional review. "Recovered" means flagged by the indicated screen and present on the USTR list; it does not mean independently verified as liable. The broad screen applies the global size and industry criteria. The French screens additionally use the indicated accounting footprint. The segment-based screen adds the report's business-segment criterion. Global sales and French subsidiary sales can differ from covered revenue under the tax law. The global-ultimate-owner variant is excluded here because the report identifies attribution problems in that variant.

2018 ownership tree, but their accounts cannot be reconstructed from the unconsolidated Orbis records. Expedia's French revenue is attributed to a sibling holding company, Liberty Expedia Holdings. The USTR list's industry codes can also be misleading: Verizon is coded as a telecommunications firm, but it is on the list for the advertising business of Verizon Media (Yahoo and AOL). In the other direction, retailers, software companies, and payments providers can meet broad size and industry criteria while earning revenue from services outside the tax base. Screening on the global threshold and a broad in-scope code alone flags groups such as Walmart and Costco.

These considerations support retaining a documented research roster rather than applying an automatic liability indicator from financial accounts. The roster therefore retains all 27 USTR groups regardless of their screen results, preserving comparability with the USTR starting list despite uncertainty arising from missing data and ambiguous taxable-service coverage. Zalando remains a potentially out-of-scope case because its measured in-scope revenue falls below the global threshold. The same considerations explain why parent revenue shares should be interpreted as measures of business composition. The shares are not estimates of the share of revenue on which a firm actually paid the French tax.

A.4 Selecting additions to the USTR roster

The financial screens identify candidates for business-model review. The final research roster retains the 27 USTR groups independently of their accounting-screen results and adds eight groups selected through two parallel screens. The 893 firms passing the broad size and industry screen therefore form a search pool for potential additions, rather than a roster that is reduced mechanically to 35 firms.

The first path uses worldwide revenue, recorded French revenue, and industry codes. Of the 893 firms passing the broad screen, 129 also pass the French revenue requirement using the operating-parent aggregation. Removing the 16 USTR groups and two duplicate representations of USTR groups—Match Group–OLD and Liberty Expedia Holdings—leaves 111 potential additions. Restricting these firms to the strict industry-code set removes 79 broad-code-only matches and leaves 32 candidates. Business-model review

classifies eight as defensible additions, nine as borderline, and 15 as primarily first-party software, IT, or payments businesses. The eight additions are Omnicom, Publicis, Interpublic, Dentsu, Cheil, TripAdvisor, ANGI, and Just Eat.

The second path uses business-segment revenue shares to narrow the activity criterion. The operating-parent segment screen flags 50 firms, including ten USTR groups. Removing those ten groups, the same two duplicate representations, and 17 firms that pass only the broader segment definition leaves 21 candidates passing the narrower segment criterion. Business-model review retains five: Omnicom, Publicis, Interpublic, TripAdvisor, and ANGI. Four candidates remain borderline and 12 primarily provide first-party software, IT, or payments services. The industry-based path recovers three additional groups—Dentsu, Cheil, and Just Eat—that lack the segment splits needed for the segment-based test. Their absence from that test reflects missing data rather than a finding that their activities fall outside the tax base.

The business-model review asks a single question of each surviving firm: does its revenue come predominantly from one of the taxed services—digital intermediation, targeted advertising based on user data, or the sale of user data—or from activities the tax excludes? Firms' descriptions of their businesses and segments in annual reports and the strict segment share inform the answer. The review sorts firms into three tiers: defensible additions, borderline cases, and mechanical passes whose codes fall in scope but whose business does not. Table A.2 lists the outcome. The exclusions reflect the distinction between an industry or segment code and the service actually supplied. Firms such as IBM, Oracle, SAP, Autodesk, ADP, and payments providers can pass a code-based screen while primarily earning revenue from their own products or services. The review also leaves mixed or ambiguous businesses outside the additions: TUI, Thomas Cook, H.I.S., and Fosun Tourism predominantly supply first-party travel services; IHS Markit and FactSet sell financial rather than user data; GfK's market-research data is not clearly user data generated on a digital interface; JCDecaux's advertising is primarily out-of-home; Adobe combines advertising-related activities with first-party software; and Comcast's in-scope advertising is a thin share of its revenue (a strict segment share of 0.12). These classifications explain why passing the accounting thresholds alone does not establish likely liability for every candidate.

Two of the eight additions, TripAdvisor and Just Eat, were already on the hand-compiled candidate list as medium-confidence intermediaries. ANGI and the five agency groups entered only through the industry-peer expansion. This is the main practical payoff of widening the universe beyond the hand-compiled candidates. Two further checks guard against missed groups. Taking the union of candidates across all five French aggregation rules adds firms whose French revenue comes from misattributed ownership-tree paths rather than from digital services. For large firms that pass the size threshold but have no resolved French-entity revenue, missing accounts may reflect genuinely zero revenue attributable to a French permanent establishment rather than a data gap. This does not, however, establish zero taxable-service revenue attributable to French users under the DST's allocation rules.

The resulting roster contains the 27 USTR groups plus five additions supported by both screening paths and three recovered through the industry-based path, for 35 groups in total. More generally, the roster

Table A.2: Business-model review of non-USTR firms passing the financial screens

Tier	Firms
Defensible additions (8)	
Intermediation	TripAdvisor (incl. Viator and TheFork), ANGI (HomeAdvisor), Just Eat
Advertising	Omnicom, Publicis, Interpublic, Dentsu, Cheil
Borderline, not added	
Travel, mostly first-party	TUI, Thomas Cook, H.I.S., Fosun Tourism
Data, not user data	IHS Markit, FactSet, GfK
Mixed or thin advertising	JCDecaux (out-of-home), Adobe, Comcast
Code-based passes outside the taxed services	
	IBM, Oracle, SAP, ADP, Fidelity National Information Services, Conduent, Open Text, Akamai, Corpay, Autodesk, Exela, WEX, and others

Notes: Firms are the union of the non-USTR survivors of the strict industry-code path (32 firms) and the strict segment-share path (21 firms) described in the text. The five advertising-agency groups fall in the same service class as Criteo on the USTR list; for agencies, only remuneration for covered targeted-advertising services enters the tax base, so total agency turnover is not a measure of taxable receipts. Tiers record a judgment about each firm's predominant business in 2018, not a verified statutory classification.

records likely rather than confirmed liability: the accounting and business-model evidence does not verify the statutory revenue base or individual tax payments. Two outside sources are consistent with a list of this size. In answer to a parliamentary question, the French tax administration reported that 27 firms paid the DST for 2019, 28 for 2020, and 36 for 2021 (Assemblée Nationale, 2023). Its unit of count is the contributing legal entity, which need not map one-for-one to a corporate group, so the totals serve as an indication of the right order of magnitude rather than as a firm-by-firm validation. Separately, a search of corporate filings and investor materials finds explicit references to French DST liabilities for Booking, ANGI, TripAdvisor, and Match (Appendix B).

A.5 Construction of the corporate revenue table

Table 2 combines two sources. Consolidated revenue and its split by activity (columns 1, 3, and 4) come from a structured extraction of each group's own FY2018 financial reports. French-entity turnover (column 2) comes from Orbis subsidiary accounts. Columns 5–8 are ratios of these four columns. Apart from a small number of documented adjustments, listed in Table A.3, every entry follows mechanically from the two sources.

Filing extraction. For each group, the documents covering its FY2018 fiscal year were collected:

- SEC filers: the annual report on Form 10-K or 20-F, the following year's report for restated comparatives, and quarterly earnings releases. Uber, Airbnb, and Wish were private in 2018, so their IPO prospectuses take the place of an annual report.
- The 12 groups that do not file with the SEC: annual reports, results releases, and investor presentations from their own websites.

In total, this amounts to 324 SEC documents and 52 home-country documents. FY2018 is the fiscal year ending between June 2018 and May 2019. Most groups report on a calendar year, but Microsoft's fiscal year ends in June 2018, Apple's in September 2018, and Alibaba's and Recruit's in March 2019.

Figures were extracted with AI assistance following a written specification. Each figure is recorded exactly as printed, with its scale, currency, document locator, and a verbatim quote of the source row. The quotes are checked automatically against the document text. An item a firm does not disclose is recorded as a disclosure gap in the firm's own words, never as a zero. Each group's total revenue is checked against Compustat, and gaps above one percent are flagged. Two such gaps reflect entity differences rather than errors. Compustat's FY2018 record under the project's identifier for Match Group is its then parent, IAC, so the table uses the standalone Match Group's filings. Cheil's English-language releases label gross profit as revenue (discussed below). Figures are taken as first reported in FY2018 documents; restated comparatives from FY2019 reports are used only where no first-reported figure exists.

Revenue classification. Each extracted revenue line is assigned to one of six categories:

- digital advertising inventory sold by the firm;
- advertising-agency fee income;
- first-party online sales of the firm's own inventory, recorded gross;
- intermediation revenue from marketplaces, travel, mobility, delivery, classifieds, jobs, and local services, recorded as net commissions and fees;
- fees of the travel distribution systems (Amadeus, Sabre, and Travelport);
- other revenue, such as hardware, software, cloud services, subscriptions, and traditional staffing.

Column 3 ("Ads") is the sum of digital advertising and agency fees. Column 4 ("Intermediation") is the sum of intermediation and distribution-system fees. First-party sales and other revenue enter neither column, so columns 3 and 4 need not sum to column 1.

Firms often print several alternative breakdowns of the same revenue—Expedia by service type and by business model, and Match Group by region and by brand—and summing across them double-counts. For each group, the revenue table therefore uses a single breakdown: the table whose lines sum most closely to total revenue. Where a firm's tables mix subtotals with their components, as with Alibaba, the lines were selected by hand. For all 35 groups, the chosen breakdown sums to within two percent of total revenue.

Exchange rates. Amounts are converted to euros using the average of Orbis's 2018 month-end exchange rates. The same calendar-2018 average applies to every group, including those whose fiscal year ends in March 2019. The cross-check against Compustat, by contrast, uses an average over each firm's own fiscal year; the difference does not affect the revenue shares, which are ratios within a firm.

French-entity turnover. Column 2 sums FY2018 turnover from the unconsolidated Orbis accounts of the French entities attributed to each group in its 2018 ownership tree. The tree is held fixed at 2018, and a reported zero is treated as missing. French entities file calendar-year accounts, so for groups whose fiscal year does not end in December the numerator and denominator of column 8 cover periods that differ by up to six months. This turnover is not sales to French users, and it is not the DST base. For most platforms the French entity is a service company that bills its parent on a cost-plus basis, while revenue from French users is booked by an Irish, Luxembourg, or U.S. principal. Thus Alphabet's, Amazon's, and Booking's French entities account for 0.4 percent or less of group revenue. Several entries need specific caveats:

- Amazon's French retail activity runs through a French branch of a Luxembourg company and is absent from these accounts.
- Criteo and Publicis are headquartered in France, so their French entities include the group parent and column 8 reflects incorporation rather than French activity.
- Amadeus's French turnover, about €7.4 million per French employee, reflects billing to the group by a research and service center rather than local sales.
- Randstad's and Recruit's French entities are staffing businesses whose turnover reflects placed temporary workers.

Five groups—Rakuten, ContextLogic, Zalando, ANGI, and TripAdvisor—have no French subsidiary accounts in Orbis and show dashes. Column 2 counts all directly attributed French entities. It is therefore related to, but not identical to, the alternative French footprints used in the liability screens of Table A.1. Column 8 divides column 2 by column 1.

Adjustments. Table A.3 lists the entries that depart from the mechanical construction, together with the reason for each.

A.6 Linking WTM domains to roster groups

WTM reports traffic by domain and carries no ownership information. Its company field identifies the operators of trackers embedded in a page, not the owner of the page. A domain-by-domain ownership classification is therefore constructed with language-model agents and researcher review. Each domain is assigned to the group that owned it in December 2018, and that assignment is held fixed over the whole sample period.

Domain universe. The universe is every domain that ranked among the top 2,000 in France, Germany, the European aggregate, the United States, or the global aggregate in any month before March 2022: 6,841 domains.

Table A.3: Adjusted entries in the corporate revenue table

Group	Adjustment and reason
Verizon	Advertising is Oath (Verizon Media) revenue of $7.7 billion, which the 10-K reports in its text rather than in a revenue table; the remainder of Verizon's revenue is telecommunications.
Snap	Advertising is approximated as 99 percent of total revenue, the share Snap reports; its filings give no separate advertising amount.
Cheil	Total revenue is taken from Orbis. Cheil's English-language releases report gross profit under the label "revenue", and its gross sales appear only in the Korean audited report. Its advertising amount is omitted because the available figure is on the gross-profit basis.
Apple, Randstad	No split is shown. Apple does not report App Store revenue separately from its other services, and Randstad does not separate digital from traditional staffing revenue.
Rakuten	Advertising revenue is embedded in mixed revenue lines and is not separately allocated, so the advertising entry is zero by construction rather than by disclosure.
Match Group	Dating subscriptions are classified as other revenue, so intermediation is zero. Subscription revenue is likely within the tax base; taking it into account would place Match's in-scope revenue above the global threshold.
Agencies	Agency fee income includes traditional as well as digital advertising services, so column 3 overstates the agencies' digital advertising revenue. Only remuneration for covered targeted-advertising services enters the tax base.

Ownership labeling. The first pass labels the 2,794 distinct domains in the December 2018 WTM release. Each domain receives an ownership label (whether a roster group owned it in December 2018, and which one) and three activity flags following the statutory definitions: digital intermediation, targeted advertising, and the sale of user data. Language-model agents labeled the domains in chunks of about 50. All agents received the same written instructions and the same reference document. The reference document sets out the statutory scope rules with in-scope and out-of-scope examples and contains a catalog of the 27 USTR groups plus TripAdvisor, listing each group's major operating domains and brands as of December 2018. Agents were instructed to recognize country-code variants and subsidiary brands not listed in the catalog, to use web searches for unfamiliar domains, to apply the definitions to what each domain did in December 2018, and never to assign a parent they could not confirm. An unconfirmed ownership was labeled non-liable with a note. The reference document flags dated ownership changes near the treatment date, including Flickr's sale by Verizon in April 2018, Recruit's acquisition of Glassdoor in June 2018, and Schibsted's ownership of its classifieds sites before the 2019 Adevinta spin-off. A script checked the combined output for complete coverage, valid values, and internal consistency, and 45 borderline rows were set aside for review.

The second pass extends the labels to the rest of the universe. Its 3,694 first-label families not covered by the first pass were represented by their most popular domain. In a triage tier, agents labeled these domains from prior knowledge without web search, in chunks of 150, and recorded their confidence in each ownership call. A domain went to a research tier with web search if the triage agent suspected or could not rule out ownership by a roster group, or if a roster-owned domain's activity was uncertain; 22 domains reached that tier. A deterministic match against the catalog's domains and country-code variants served as a

backstop for missed ownership. Labels were then propagated from each family's representative to its other domains, and a stratified sample of 42 rows was checked by hand.

The two passes assign 363 domains to 26 of the 28 catalogued groups. Amadeus and Travelport, both business-to-business distribution systems, own no domain in the universe. The labeling catalog predates two roster additions that operate consumer websites, ANGI and Just Eat. Their domains were assigned by hand: `homeadvisor.com` to ANGI, and `just-eat.co.uk` and `just-eat.fr` to the standalone Just Eat plc that existed in 2018. The Takeaway.com brands (Lieferando, Thuisbezorgd, Pyszne) were a separate firm in 2018 and are not assigned to Just Eat. The five advertising agencies, the remaining additions, operate no consumer website and own no domain in the universe. Thus 28 of the 35 roster groups own at least one domain in the universe, and seven (the five agencies, Amadeus, and Travelport) own none.

Committee review. Before any estimates were produced, a committee of language-model agents reclassified a 2,020-domain traffic universe: the 1,854 domains observed in both the French and global WTM panels in at least three months of April–November 2018 and at least six post-announcement months, plus every roster-owned domain observed at all. Three agents classified each domain independently. The first classified the product and the domain's role, distinguishing consumer destinations from infrastructure, static-content, authentication, developer, advertising-technology, and enterprise domains. The second classified the revenue model and scored exposure to each taxed activity. The third assessed competitive closeness to the roster groups' services. Agents saw only pre-treatment or time-invariant context, and batches kept all domains of a family together. An adjudicating model merged the three classifications field by field, and an auditing model re-examined 766 rows: every proposed low-exposure control, every row with an ownership or family-collision flag, and every row flagged for review. The researcher decided the 24 rows that remained unresolved.

The review changed the treated set in five places. It added two domains the labels had missed: `tradera.com`, flagged as an eBay subsidiary, and `dndbeyond.com`, a Curse Media property assigned to Amazon on the same basis as Futhead, Gamepedia, and Hearthpwn. It removed three domains: `amazonbrowserapp.de`, `fonts.googleapis.com`, and `uberinternal.com`. It also corrected the owner of `avito.ru` to Naspers, which controlled it in 2018, so that domain is not treated. The classifications were then frozen, and every later change is recorded in a changelog.

Website families. A website family pools a brand's domains under its first domain label, so that `google.fr`, `google.de`, and `google.com` form one family. Pooling neutralizes traffic moves between country domains and handles brands whose French storefront is a separate domain. Family traffic in a month is the sum of popularity over the member domains observed that month, computed before taking logs.

Result. The linkage assigns 365 domains in 222 website families to 28 roster groups. These families are the starting point for the sample filters below.

A.7 From linked families to the French liable sample

Five filters take the 222 linked families to the 55 families in the French sample. Table A.4 reports the families and groups that remain after each filter. The paragraphs below state each filter, its purpose, and the families and groups it removes. Families are named by their first domain label.

Table A.4: From roster-linked website families to the French liable sample

Step	Families	Groups	Families dropped	Groups lost
Linked families	222	28		
1. Curated family roster	219	28	3	0
2. At least one consumer-facing domain	185	25	34	3
3. Any French traffic in April–November 2018	131	22	54	3
4. At least four French months in April–November 2018	76	21	55	1
5. Activity review	55	20	21	1

Notes: Families pool domains under their first domain label. Groups are the roster groups owning at least one remaining family. Filter 4 also removes PriceMinister, which has four or more French months; see the text. The 55 families are listed in Table 3.

Filter 1: curated family roster (222 to 219). Each family must have a single, verified December 2018 owner and correspond to a commercial property of that owner. The filter removes three families. `avito` (Schibsted): Schibsted's Moroccan site `avito.ma` shares its first label with `avito.ru`, which Naspers controlled, so the family would pool a liable and a non-liable owner. `learndigital` (Alphabet): `learndigital.withgoogle.com` is Google's free digital-skills training program, carries no in-scope revenue, and its family key is a subdomain rather than a brand. `tradera` (eBay): its December 2018 ownership is disputed. No group is lost.

Filter 2: at least one consumer-facing domain (219 to 185). The outcome is consumer traffic, so a family must contain at least one domain that consumers visit as a destination. Domains classified as infrastructure, content delivery, authentication, developer tools, advertising technology, enterprise software, or corporate pages measure server or business activity rather than consumer demand. A family is removed if every member domain is so classified. Ten such families with French traffic before the announcement are instead passed to the activity review (Filter 5), so that their treatment is an explicit decision: `android`, `github`, `googleusercontent`, `microsoftonline`, `oath`, `sharepoint`, `visualstudio`, `futhead`, `gamepedia`, and `hearthpwn`. The filter removes 34 families:

- Microsoft (7): `azure`, `dynamics`, `lync`, `microsoftstream`, `windows`, `windowsazure`, `yammer`.

- Alphabet (5): doubleclick, googlesyndication, material, nest, virustotal.
- Amazon (5): alexa, amazonaws, dndbeyond, ring, ssl-images-amazon.
- Verizon (3): huffpost, verizon, yimg.
- Alibaba (2): alicdn, aliyun. Meta (2): oculus, workplace. Sabre (2): getthere, synxis. Twitter (2): periscope, twimg.
- One each: Criteo (criteo), eBay (ebaystatic), Expedia (expediapartnercentral), Rakuten (rakuten-sec), Snap (snapchat), TripAdvisor (seatguru).

Criteo, Sabre, and Snap lose all their families.

Filter 3: any French traffic in April–November 2018 (185 to 131). Measuring a response to the tax requires French traffic before the announcement. A family is retained if any member domain has positive French popularity in at least one month of April–November 2018, the pre-announcement months covered by the French series. WTM publishes only the top sites in each country and month, and its French list was short before the announcement: about 600–875 sites in April–August 2018 and 2,000–2,400 in September–November 2018, rising to 6,000 from August 2019. The families removed here first appear in the French list between December 2018 and June 2020. They were below the reporting cutoff in 2018, not without French users. The filter removes 54 families:

- Amazon (7): 6pm, abebooks, bookdepository, boxofficemojo, shopbop, zappos, zvab.
- Schibsted (7): aftenposten, avendrealouer, coches, e24, hasznaltauto, svd, videdressing.
- Expedia (5): abritel, homeaway, orbitz, travelocity, trivago.
- Microsoft (5): groupme, mixer, mojang, outlook, powerbi.
- eBay (4): bilbasen, brands4friends, ebaystores, stubhub.
- Axel Springer (3): bz-berlin, stylebook, totaljobs. Match Group (3): disonsdemain, lovescout24, ourtime. Verizon (3): autoblog, mapquest, rivals.
- Alphabet (2): tenor, waze. Booking (2): opentable, priceline. Rakuten (2): ebates, kobo. Randstad (2): military, monster. Recruit (2): jalan, suumo. TripAdvisor (2): cruisecritic, viator.
- One each: Alibaba (trendyol), ANGI (homeadvisor), Just Eat (just-eat), Uber (ubereats), Zalando (zalando-lounge).

Randstad, ANGI, and Just Eat lose all their families.

Filter 4: at least four French months in April–November 2018 (131 to 76). Estimating a family's response requires a pre-announcement baseline of several months. A family is retained if it has positive French popularity in at least four of the eight months of April–November 2018. Of the 54 families that fail, 39 have three French months and 15 have one or two. The 49 of them listed in Table 3 all first appear in the French list in September–November 2018, when the list deepened. The 54 families are:

- Axel Springer (13): `autobild`, `businessinsider`, `caradisiac`, `finanzen`, `idealo`, `immonet`, `immoweb`, `immowelt`, `kaufda`, `meinestadt`, `stepstone`, `transfermarkt`, `travelbook`.
- Schibsted (11): `aftonbladet`, `blocket`, `finn`, `infojobs`, `jofogas`, `kufar`, `olx`, `shpock`, `tori`, `vg`, `willhaben`.
- Alibaba (4): `1688`, `lazada`, `tmall`, `youku`. Amazon (4): `audible`, `mturk`, `souq`, `woot`. Booking (4): `agoda`, `kayak`, `momondo`, `swoodoo`. eBay (4): `2dehands`, `dba`, `gumtree`, `motor-talk`.
- Expedia (3): `expedia`, `fewo-direkt`, `vrbo`. Rakuten (3): `infoseek`, `overdrive`, `rakuten`.
- Match Group (2): `match`, `tinder`. Verizon (2): `engadget`, `techcrunch`.
- One each: Alphabet (`kaggle`), Groupon (`groupon`), Microsoft (`lynda`), Recruit (`glassdoor`).

The filter also removes PriceMinister (Rakuten). Its French traffic falls by three orders of magnitude in September 2018, leaving no usable September–November 2018 baseline. Groupon loses its only family. Forty-nine of the removed families pass the activity review and are listed in Table 3 as roster websites with limited pre-treatment data.

Filter 5: activity review (76 to 55). The tax applies to revenue from targeted advertising and digital intermediation. Websites that sell the group's own products, provide communications services, or serve as infrastructure do not carry the taxed activity, so their traffic does not measure a response to the tax. Each remaining website was hand-coded as of 2018, using AI-assisted drafts confirmed by the researcher site by site, under a written codebook:

- *Targeted advertising:* paid third-party placements selected using user information, such as interest, behavioral, location, or search-query targeting, or sponsored listings.
- *Paid digital intermediation:* monetized user-to-user connections, such as commissions, listing or access fees, and paid matching features.
- *Materiality:* ancillary features are coded as absent.

Each activity is rated none, weak, or material, and a website is retained if either activity is material. Each row records its caveats and selected sources, such as annual reports, archived snapshots of the site, and

media kits. The ratings are qualitative judgments about a site's dominant activity, not measured revenue shares. The review removes 21 families:

- Microsoft (10): `github`, `microsoft`, `microsoftonline`, `minecraft`, `office`, `office365`, `sharepoint`, `skype`, `visualstudio`, `xbox`.
- Alphabet (3): `android`, `blogger`, `googleusercontent`.
- Apple (2): `apple`, `icloud`. Meta (2): `messenger`, `whatsapp`. Verizon (2): `oath`, `verizonwireless`.
- One each: Amazon (`primevideo`), Twitter (`pscp`, Periscope).

Of the ten families passed through Filter 2, the three Curse network sites (`futhead`, `gamepedia`, `hearthpwn`) are retained as ad-supported, and the seven infrastructure families are removed. Apple loses all its families.

Result. The French sample contains 55 families belonging to 20 groups, listed in Table 3. Available January–December 2019 observations enter the traffic panel without a complete post-announcement coverage requirement. Annual estimates compare September–November 2018 with January–December 2019, excluding December 2018.

Fifteen of the 35 roster groups contribute no family to the sample: the five advertising agencies, Amadeus, and Travelport own no domain in the universe; Criteo, Sabre, and Snap own only non-consumer domains (Filter 2); Randstad, ANGI, and Just Eat have no French traffic before the announcement (Filter 3); Groupon has too few French months (Filter 4); and Apple's sites are first-party products and services (Filter 5). Absence from the sample is not evidence of non-liability.

Nonliable comparison families. Comparison families are built from the same labeled universe. They are domains labeled as not owned by a roster group, excluding Wikipedia, which serves as the normalizing site. These domains are pooled into families by first label in the same way. A comparison domain whose first label coincides with a liable family keeps its full domain as its key, so that, for example, Yahoo Japan stays separate from Verizon's Yahoo. The 638 comparison families are those observed in all six months of June–November 2018. They satisfy the empirical coverage requirements and are distinguished by observed advertising exposure; the design does not treat all other websites in the underlying traffic universe as unexposed controls.

The main advertising-composition analysis retains all 55 liable families and measures parent revenue composition using fiscal-year 2018 accounts. The 35-group institutional roster and the 28-firm disclosure inventory in Appendix B serve different purposes from the traffic sample, as Table A.5 summarizes.

Table A.5: Distinct populations used in the institutional and empirical work

Population	Firms or parents	Website families
Institutional research roster	35	–
Disclosure inventory, including home-country firms	28	–
SEC-filer component of disclosure inventory	21	–
Roster groups linked to websites in the WTM universe	28	222
French liable traffic sample	20	55
Main advertising-composition sample	20	55

Notes: These are populations with different coverage requirements, not a sequence of nested sample restrictions. Disclosure counts refer to firms represented in the inventory, not to firms with an available filing in every year. The linked-websites row counts first-label website families after the committee review and before any sample filter. The final two rows describe the same liable-family roster used in the headline and main advertising-composition analyses. No aggregate traffic outcome is imputed to groups lacking observed websites.

B Corporate disclosures and tax collections

B.1 Sources and coverage

The disclosure inventory contains 28 firms, including 21 SEC filers and seven firms covered through home-country reports. It records fiscal years 2019–2024, with the initial review also consulting 2018 filings. SEC materials include annual and quarterly reports and earnings releases; the narrative search covers risk factors, management discussion, legal proceedings, and financial-statement notes. For each firm-year, the inventory records whether a digital services tax is mentioned, whether France is identified, whether a numerical amount is supplied, and what that amount measures.

Reported amounts include expenses, changes in broader expense categories, payments, and contingent exposures. The inventory records their accounting meaning and geographic coverage separately. An amount covering several countries can be assigned to France only when the report provides a country breakdown.

Coverage varies with listings, delistings, ownership changes, and the availability and parsing of reports. A missing France-specific amount means that the review did not recover one; it provides no evidence of zero liability or payment.

B.2 DST expense disclosures

Table B.1 presents four 2019 disclosures recorded in the review. Booking identifies $36 million associated with French digital service taxes in its discussion of general and administrative expenses. ANGI identifies $0.9 million in its European expense discussion. TripAdvisor records a $3 million estimate immediately after describing the French law. The Match segment's $4.7 million figure, reported in IAC's consolidated filing, is an increase in non-income taxes primarily related to the French tax; it is not a separately identified French tax total (Booking Holdings, 2020; ANGI Homeservices, 2020; TripAdvisor, 2020; IAC/InterActiveCorp, 2020).

Table B.1: Selected 2019 disclosures and their accounting meaning

Firm or segment	Reported amount	Interpretation
Booking Holdings	$36m	French digital service taxes identified in the general and administrative expense discussion.
ANGI	$0.9m	Amount associated with the French tax in the increase in European general and administrative expenses.
TripAdvisor	$3m	Estimated digital service tax expense disclosed in the discussion of France's 2019 law.
Match segment (IAC)	$4.7m	Increase in non-income taxes primarily related to the French tax; the entire amount is not isolated as French DST.

Notes: Amounts are in the reporting currency. They are expense disclosures, not verified cash payments. Source: FY2019 Forms 10-K, management discussion, for Booking Holdings (2020); ANGI Homeservices (2020); TripAdvisor (2020); IAC/InterActiveCorp (2020). The Match observation is located in IAC's discussion of its MTCH segment.

Adevinta's 2022 annual report provides a country-specific example outside the SEC inventory. It describes payments on account of €11 million for 2022 and €22 million for 2019–2021 following a contested French ruling. The company appealed and expected to pay an additional €6 million in early 2023 (Adevinta, 2023). Its €33 million payment during 2022 thus covered several tax years while the dispute remained unresolved.

The four 2019 disclosures appear in discussions of operating expenses, where a search confined to the corporate income-tax footnote would miss them. They document costs associated with the tax but reveal little about changes in spending on particular services or geographic markets.

B.3 Government receipts

Government receipts show the scale of collections. A parliamentary response reports €277 million from 27 contributing enterprises in 2019, €375 million from 28 in 2020, and €474 million from 36 in 2021 (Assemblée Nationale, 2023). Table B.2 reproduces these totals. The response classifies nationality by the contributing enterprise's registered office, so its enterprise counts cannot be matched directly to the ultimate parents in the firm list.

Table B.2: French tax receipts and contributing enterprises, 2019–2021

Calendar year	Receipts (€m)	Contributing enterprises
2019	277	27
2020	375	28
2021	474	36

Notes: Source: DGFiP data reported in the government's February 14, 2023 response to parliamentary question 2974 (Assemblée Nationale, 2023). Receipts concern budget line 1430. The administrative enterprise count is not assumed to equal a count of ultimate-parent groups.

Collections and reported corporate expenses cover different firms, periods, and accounting concepts. Comparing their totals therefore cannot establish how much of the tax appears in public filings.

C Advertising demand and tax responses

This appendix derives advertising demand and the tax responses for the content-consumption benchmark in Section 3.1.

C.1 Downstream demand for advertising

This section provides a derivation for downward-sloping demand. The assumptions are explicit: advertising helps firms reach customers, impression requirements increase with sales, and underlying product valuations remain unchanged. This approach emphasizes the idea that what social media platforms are doing is supplying an input into production of a final good.

Consider a competitive downstream product market. Let q denote product sales, $P_B(q)$ inverse consumer demand, and $P_S(q)$ marginal supply cost excluding advertising, with $P_B'(q) < 0$ and $P_S'(q) > 0$. Advertising is a required selling input that allows firms to reach customers. It does not change consumers' underlying product valuations.

Supporting sales q requires $A = \kappa(q)$ advertising impressions, where

$$\kappa(0) = 0, \qquad \kappa'(q) > 0, \qquad \kappa''(q) \geq 0.$$

The linear requirement $\kappa(q) = \bar{\kappa}q$, with $\bar{\kappa} > 0$, is a special case. At price p_A per impression, total downstream supply cost is

$$\int_0^q P_S(x)\,dx + p_A\kappa(q).$$

Product-market equilibrium therefore satisfies

$$P_B(q) = P_S(q) + p_A\kappa'(q).$$

Advertising enters the marginal cost of supplying the product. Since $\kappa'(q) > 0$, the requirement can be inverted to obtain $q = \kappa^{-1}(A)$. Inverse advertising demand is then

$$P_A(A) = \frac{P_B\big(\kappa^{-1}(A)\big) - P_S\big(\kappa^{-1}(A)\big)}{\kappa'(\kappa^{-1}(A))}.$$

Writing $q = \kappa^{-1}(A)$, differentiation gives

$$P_A'(A) = \frac{[P_B'(q) - P_S'(q)]\,\kappa'(q) - [P_B(q) - P_S(q)]\,\kappa''(q)}{[\kappa'(q)]^3} < 0$$

on the positive-price domain $P_B(q) > P_S(q)$. Advertising expenditure is determined by this market equilibrium, rather than by an exogenous advertising budget. Assume P_B and P_S are twice continuously differentiable and κ is three times continuously differentiable, so P_A is twice continuously differentiable.

C.2 Conditional expenditure and impression volume

Assume that $\mathcal{L}$ and F are twice continuously differentiable and that the conditional minima defining $U(A)$ exist and vary smoothly on the relevant positive interior branch. For a fixed volume A, the conditional expenditure problem has Lagrangian

$$\mathcal{J} = g_U + \lambda \left[A - a\mathcal{L}(a, g_U) \right],$$

where λ is the multiplier on the impression requirement. Its first-order conditions imply

$$\lambda = \frac{1}{a\mathcal{L}_{g_U}} > 0, \qquad \mathcal{L} + a\mathcal{L}_a = 0.$$

The envelope theorem gives $U'(A) = \lambda$. Hence total marginal expenditure is positive:

$$C'_A(A) = (F^{-1})'(A) + \frac{1}{a\mathcal{L}_{g_U}} > 0.$$

Holding A fixed, the constraint locally defines the required g_U as a function of a. At its stationary point, the second derivative of this required expenditure is

$$\left.\frac{\partial^2 g_U}{\partial a^2}\right|_A = -\frac{2\mathcal{L}_a + a\mathcal{L}_{aa}}{a\mathcal{L}_{g_U}} > 0.$$

Thus the strict conditional second-order condition requires $2\mathcal{L}_a + a\mathcal{L}_{aa} < 0$.

For reduced-form profit $\Pi(A;\tau) = (1-\tau)P_A(A)A - C_A(A)$, the first-order and strict second-order conditions are

$$(1-\tau)\left[P_A(A) + AP'_A(A)\right] = C'_A(A),$$

and

$$\Pi_{AA}(A;\tau) = (1-\tau)\left[2P'_A(A) + AP''_A(A)\right] - C''_A(A) < 0.$$

Implicit differentiation yields

$$\frac{dA}{d\tau} = \frac{P_A(A) + AP'_A(A)}{(1-\tau)\left[2P'_A(A) + AP''_A(A)\right] - C''_A(A)} < 0.$$

The numerator equals $C'_A(A)/(1-\tau) > 0$. The expenditure and price responses follow from the production constraint, conditional expenditure function, and advertising demand:

$$\frac{dg_A}{d\tau} = (F^{-1})'(A)\frac{dA}{d\tau} < 0,$$
$$\frac{dg_U}{d\tau} = \frac{1}{a\mathcal{L}_{g_U}}\frac{dA}{d\tau} < 0,$$
$$\frac{dp_A}{d\tau} = P'_A(A)\frac{dA}{d\tau} > 0.$$

C.3 Advertising intensity and traffic

The possibility of a traffic increase depends on how advertising intensity and user-facing expenditure interact in the traffic function. In the multiplicatively separable case $\mathcal{L}(a, g_U) = H(g_U)V(a)$, optimal intensity is locally unchanged by the tax and traffic falls. A positive traffic response therefore requires a departure from this separable benchmark, with intensity falling sufficiently to offset the loss from lower user-facing expenditure.

Differentiating the conditional intensity condition gives

$$(2\mathcal{L}_a + a\mathcal{L}_{aa})\frac{da}{d\tau} + (\mathcal{L}_{g_U} + a\mathcal{L}_{ag_U})\frac{dg_U}{d\tau} = 0.$$

Since $dg_U/d\tau < 0$, intensity falls, rises, or is locally unchanged according to whether $\mathcal{L}_{g_U} + a\mathcal{L}_{ag_U}$ is positive, negative, or zero. The interior condition also implies $\mathcal{L}_a = -\mathcal{L}/a < 0$; this benchmark does not cover an interior optimum at which advertising increases traffic.

Substituting the intensity and expenditure responses into $dL/d\tau = \mathcal{L}_a(da/d\tau) + \mathcal{L}_{g_U}(dg_U/d\tau)$ yields

$$\frac{dL}{d\tau} = \frac{\mathcal{L}_{g_U}(\mathcal{L}_a + a\mathcal{L}_{aa}) - a\mathcal{L}_a\mathcal{L}_{ag_U}}{a\mathcal{L}_{g_U}(2\mathcal{L}_a + a\mathcal{L}_{aa})}\frac{dA}{d\tau}.$$

The denominator and $dA/d\tau$ are both negative. Using $\mathcal{L}_a = -\mathcal{L}/a$, traffic therefore rises precisely when

$$\frac{a^2\mathcal{L}_{aa}}{\mathcal{L}} + \frac{a\mathcal{L}_{ag_U}}{\mathcal{L}_{g_U}} > 1.$$

Traffic falls when the left-hand side is below one and is locally unchanged when it equals one. Equivalently, $L = A/a$ implies

$$\frac{d\log L}{d\tau} = \frac{d\log A}{d\tau} - \frac{d\log a}{d\tau}.$$

Since impressions fall, traffic can rise only if advertising intensity falls proportionately more than impressions.

For multiplicatively separable traffic, $\mathcal{L}(a, g_U) = H(g_U)V(a)$, with $H, V > 0$ on the relevant domain,

the intensity condition becomes

$$V(a) + aV'(a) = 0.$$

The strict conditional second-order condition selects a locally isolated solution independent of g_U. Thus $da/d\tau = 0$, and lower user-facing expenditure reduces traffic. All comparative statics in this appendix are evaluated at the maintained interior optimum.

D Intermediation costs, fees, and traffic

This appendix derives the tax responses for the transaction-intermediation model in Section 3.2. Intermediation expenditure is the operating cost of facilitating transactions. It does not independently shift buyer demand, seller supply, or service quality.

D.1 Transaction volume and operating expenditure

Assume that inverse buyer demand D^{-1}, inverse seller supply S^{-1}, and the transaction technology F are twice continuously differentiable on the relevant domains. The fee schedule and required expenditure are

$$P_Q(Q) = D^{-1}(Q) - S^{-1}(Q), \qquad C_Q(Q) = F^{-1}(Q).$$

The demand and supply slopes imply $P'_Q(Q) < 0$. Since $F' > 0$ and $F'' \leq 0$, required expenditure has positive marginal cost and is weakly convex:

$$C'_Q(Q) = \frac{1}{F'(g_Q)} > 0, \qquad C''_Q(Q) = -\frac{F''(g_Q)}{[F'(g_Q)]^3} \geq 0, \qquad g_Q = F^{-1}(Q).$$

The platform chooses feasible transaction volumes with nonnegative fees. At a positive interior optimum of equation (9), the first-order condition is

$$(1-\tau)\left[P_Q(Q) + QP'_Q(Q)\right] = C'_Q(Q).$$

The strict second-order condition requires

$$\Pi_{QQ}(Q;\tau) = (1-\tau)\left[2P'_Q(Q) + QP''_Q(Q)\right] - C''_Q(Q) < 0.$$

Differentiating the first-order condition gives

$$\frac{dQ}{d\tau} = \frac{P_Q(Q) + QP'_Q(Q)}{(1-\tau)\left[2P'_Q(Q) + QP''_Q(Q)\right] - C''_Q(Q)} < 0.$$

The numerator equals $C'_Q(Q)/(1-\tau) > 0$, and the denominator is negative by the second-order condition. The reduction in transaction volume lowers required operating expenditure:

$$\frac{dg_Q}{d\tau} = C'_Q(Q)\frac{dQ}{d\tau} < 0.$$

D.2 Fees, prices, and traffic

Along the fee schedule, buyer demand, and seller supply, the price responses are

$$\begin{aligned}\frac{dp_Q}{d\tau} &= P_Q'(Q)\frac{dQ}{d\tau} > 0,\\ \frac{dp_B}{d\tau} &= (D^{-1})'(Q)\frac{dQ}{d\tau} > 0,\\ \frac{dp_S}{d\tau} &= (S^{-1})'(Q)\frac{dQ}{d\tau} < 0.\end{aligned}$$

The buyer-price increase and seller-compensation reduction sum to the fee increase: $dp_Q/d\tau = dp_B/d\tau - dp_S/d\tau$. Market clearing gives $S(p_S) = Q$, so the change in supply is

$$\frac{dS(p_S)}{d\tau} = S'(p_S)\frac{dp_S}{d\tau} = \frac{dQ}{d\tau} < 0.$$

Seller availability is distinct from aggregate supply and completed transactions. As in the main text, let $N = H(p_S)$ denote the number of active sellers, where H is continuously differentiable and $H'(p_S) > 0$. This participation assumption is additional to upward-sloping aggregate supply. Lower compensation then reduces seller participation:

$$\frac{dN}{d\tau} = H'(p_S)\frac{dp_S}{d\tau} = H'(p_S)(S^{-1})'(Q)\frac{dQ}{d\tau} < 0.$$

For a differentiable traffic function $L = \widetilde{\mathcal{L}}(p_B, N)$, the equilibrium traffic response is

$$\begin{aligned}\frac{dL}{d\tau} &= \widetilde{\mathcal{L}}_{p_B}\frac{dp_B}{d\tau} + \widetilde{\mathcal{L}}_N\frac{dN}{d\tau}\\ &= \left[\widetilde{\mathcal{L}}_{p_B}(D^{-1})'(Q) + \widetilde{\mathcal{L}}_N H'(p_S)(S^{-1})'(Q)\right]\frac{dQ}{d\tau}.\end{aligned}$$

Because $dQ/d\tau < 0$, traffic falls precisely when the bracketed expression is positive. The maintained assumptions $\widetilde{\mathcal{L}}_{p_B} < 0$ and $\widetilde{\mathcal{L}}_N > 0$, together with downward-sloping buyer demand, upward-sloping seller supply, and $H' > 0$, are sufficient for this result. Higher buyer prices discourage visits, while fewer active sellers reduce the availability of offers that attracts users. If the bracket is negative, traffic rises despite fewer completed transactions; if it is zero, traffic is locally unchanged. All responses are evaluated at the maintained interior optimum.

E Website-level traffic effects

Table E.1 reports annual traffic-effect estimates for each of the 55 liable website families belonging to 20 corporate parents. For each family, it presents a raw synthetic-control contrast, the composition model's prediction, and an empirical Bayes estimate combining the two. All three use the main French-only synthetic-control specification, with donors drawn from the 159 nonliable families without observed pre-announcement advertising connections to liable platforms.

Let $g_{it} = y_{it} - \widehat{y}^0_{it}$ denote family i's observed-minus-synthetic log traffic gap. The raw annual contrast is

$$\widehat{\delta}_i = \frac{1}{12}\sum_{t\in\mathcal{Q}} g_{it} - \frac{1}{3}\sum_{t\in\mathcal{P}} g_{it}, \qquad \mathcal{P} = \{\text{Sep.–Nov. 2018}\}, \quad \mathcal{Q} = \{\text{Jan.–Dec. 2019}\}.$$

December 2018 is excluded. For each point estimate, synthetic weights are fitted using available June–November 2018 French traffic and applied to the post-announcement path. The full-pipeline inference procedure refits those weights in every bootstrap draw.

The model-implied effect is

$$m_i = \widehat{\beta} + \widehat{\theta}_C \frac{C_{j(i)} - \overline{C}}{s_C},$$

where $C_{j(i)}$ is advertising's share of advertising plus intermediation revenue for family i's parent. The mean $\overline{C}$ and standard deviation s_C are calculated across the 55 liable families. The coefficients come from the original stacked composition regression. Families with the same composition therefore share the same model prediction, while their raw and EB estimates can differ.

The EB estimate is

$$\widetilde{\delta}_i = r_i\widehat{\delta}_i + (1 - r_i)m_i, \qquad r_i = \frac{\widehat{\tau}^2}{\widehat{\tau}^2 + \widehat{s}_i^2}.$$

The raw contrast's sampling variance $\widehat{s}_i^2$ is estimated using a website-specific baseline and constant 2019 shift, with a common AR(1) residual correlation and website-specific scales. Holding the original-model predictions fixed, the residual signal variance $\widehat{\tau}^2$ is estimated by marginal likelihood. The reliability weight r_i determines how strongly the raw contrast is pulled toward its composition-model prediction. It is distinct from the synthetic donor weights: larger r_i gives greater weight to the raw website contrast.

The table reports $100\widehat{\delta}_i$, $100m_i$ and $100\widetilde{\delta}_i$ in log percentage points, and $100r_i$ in the final column. It groups families by their parent's composition: pure advertising ($C = 1$; 13 families), mixed ($0 < C < 1$; 37 families), and pure intermediation ($C = 0$; five families). These are annual point estimates, calculated directly from the pre/post contrast. They need not equal averages of the separately fitted monthly EB estimates.

Table E.1: Annual website-level raw, model-implied and empirical Bayes traffic effects

Website family	Corporate parent	Ad share (%)	Raw synthetic	Model-implied	Empirical Bayes	Weight on raw (%)
Pure advertising (13 families)						
google	Alphabet (Google)	100.0	+0.48	+16.48	+3.91	78.6
youtube	Alphabet (Google)	100.0	+17.31	+16.48	+17.19	85.2
meetic	Match Group	100.0	+23.85	+16.48	+20.54	55.2
okcupid	Match Group	100.0	+88.18	+16.48	+32.44	22.3
pof	Match Group	100.0	+2.54	+16.48	+15.44	7.4
twoo	Match Group	100.0	+38.37	+16.48	+18.79	10.6
facebook	Meta (Facebook)	100.0	−2.60	+16.48	−0.12	87.0
instagram	Meta (Facebook)	100.0	+41.19	+16.48	+32.01	62.9
twitter	Twitter	100.0	+40.66	+16.48	+32.26	65.3
aol	Verizon (Verizon Media)	100.0	−3.78	+16.48	+4.83	57.5
huffingtonpost	Verizon (Verizon Media)	100.0	+20.41	+16.48	+19.26	70.6
tumblr	Verizon (Verizon Media)	100.0	+20.80	+16.48	+18.78	53.2
yahoo	Verizon (Verizon Media)	100.0	+1.89	+16.48	+4.37	83.0
Mixed (37 families)						
alibaba	Alibaba	59.1	+15.93	+2.57	+6.19	27.1
aliexpress	Alibaba	59.1	+12.24	+2.57	+5.26	27.8
taobao	Alibaba	59.1	−15.50	+2.57	+0.49	11.5
amazon	Amazon	19.1	+12.67	−11.00	−4.50	27.4
curseforge	Amazon	19.1	+17.23	−11.00	−6.38	16.4
dpreview	Amazon	19.1	+8.54	−11.00	−4.14	35.1
futhead	Amazon	19.1	−99.31	−11.00	−15.89	5.5
gamepedia	Amazon	19.1	+33.74	−11.00	+1.94	28.9
goodreads	Amazon	19.1	−4.03	−11.00	−8.91	30.0
hearthpwn	Amazon	19.1	+6.13	−11.00	−10.54	2.7
imdb	Amazon	19.1	+28.74	−11.00	+20.47	79.2
twitch	Amazon	19.1	+46.31	−11.00	+17.86	50.4
bild	Axel Springer	25.6	+0.70	−8.78	−7.42	14.4
computerbild	Axel Springer	25.6	−134.17	−8.78	−13.15	3.5
lacentrale	Axel Springer	25.6	+3.32	−8.78	−0.67	67.1
logic-immo	Axel Springer	25.6	+12.83	−8.78	+0.48	42.8
seloger	Axel Springer	25.6	−6.06	−8.78	−6.80	72.9

Continued on next page

Table E.1 (continued)

Website family	Corporate parent	Ad share (%)	Raw synthetic	Model-implied	Empirical Bayes	Weight on raw (%)
welt	Axel Springer	25.6	−44.39	−8.78	−15.65	19.3
booking	Booking Holdings	7.3	+11.31	−15.02	+0.56	59.2
rentalcars	Booking Holdings	7.3	+7.03	−15.02	−10.95	18.4
hotels	Expedia	9.7	+18.03	−14.19	−0.69	41.9
bing	Microsoft	57.1	−2.28	+1.92	+1.15	18.4
linkedin	Microsoft	57.1	+16.67	+1.92	+7.92	40.7
live	Microsoft	57.1	−9.65	+1.92	−4.11	52.2
msn	Microsoft	57.1	−21.65	+1.92	−3.89	24.7
slideshare	Microsoft	57.1	−23.07	+1.92	−6.39	33.2
leboncoin	Schibsted	37.3	−25.81	−4.81	−10.12	25.3
milanuncios	Schibsted	37.3	−16.67	−4.81	−5.41	5.1
subito	Schibsted	37.3	−24.14	−4.81	−7.03	11.5
tripadvisor	TripAdvisor (borderline)	71.6	−0.02	+6.84	+4.31	36.9
wish	Wish (ContextLogic)	12.4	−68.36	−13.27	−35.83	41.0
2ememain	eBay	11.4	−35.10	−13.62	−17.37	17.4
ebay	eBay	11.4	+16.63	−13.62	+4.92	61.3
ebay-kleinanzeigen	eBay	11.4	−50.57	−13.62	−18.89	14.3
kijiji	eBay	11.4	−11.20	−13.62	−13.31	12.9
marktplaats	eBay	11.4	−80.02	−13.62	−25.95	18.6
mobile	eBay	11.4	−64.83	−13.62	−18.01	8.6
Pure intermediation (5 families)						
airbnb	Airbnb	0.0	+0.70	−17.50	−9.55	43.7
viki	Rakuten	0.0	+11.20	−17.50	−9.59	27.5
indeed	Recruit Holdings	0.0	+7.70	−17.50	−8.11	37.2
uber	Uber	0.0	+5.53	−17.50	−12.96	19.7
zalando	Zalando	0.0	−0.03	−17.50	−15.44	11.8

Notes: Advertising share is $100C$. The three effect columns report log percentage points, without exponentiation; the final column reports the percentage weight on the raw contrast in the EB estimate. The displayed point estimates use the original fitted synthetic paths. The AR(1) noise calibration is a working within-website variance model; its formula does not add a separate donor-weight variance component. In the full-pipeline bootstrap, synthetic weights, model centers and this calibration are all reestimated, so the reported group inference incorporates fitted-weight and shared-donor variation. The table reports point estimates; reliability weights describe shrinkage and are not confidence levels.

F Supplementary heterogeneity analyses

The comparison groups follow the main-text exposure definition. Exposure requires a positive French-panel detection during September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The residual group includes unknown exposure. The 638 nonliable families divide into 469 exposed and 169 residual families.

Bootstrap inference for supplementary specifications. All standard errors and plotted intervals in this appendix use 1,999 draws with the same ordinary parent and exposure-stratified nonliable-family resampling as the main regression exhibits. Each draw retains full histories and original missingness, reestimates donor-change scales and baseline offsets, and fits synthetic weights on available June–November 2018 French observations. The French-share specifications repeat the original $\kappa = 1$ share-gap penalty and rebuild each synthetic member's weighted donor-share regressor; the joint revenue-share specification instead refits the headline traffic-only weights without this penalty. Donor copies receive separate simplex coordinates and ridge penalties, with shared multiplicities across all targets and pools. Original 55-family standardization, French-share calibration, exposure labels and eligibility rules remain fixed. Both annual specifications and primary monthly regressions are reestimated. Standard errors use the draw standard deviation, and plotted intervals are pointwise bootstrap percentiles. Appendix H reports the conditional results.

F.1 French share of global traffic

In addition to the baseline regression, the French-share specification examines whether traffic responses vary with French audience concentration. Equation 2 motivates this comparison: the effective marginal tax rate depends on French presence and French advertising prices relative to the worldwide average. Audience concentration is measured using family-level French and global WTM popularity over September–November 2018. To account for separate geographic normalization, the calibration multiplies each month's French popularity by 0.9 times Amazon.fr's global-to-French popularity ratio, sums across available months and divides by the family's global popularity summed over the same months. This calibration assumes Amazon.fr obtains 90% of its traffic from France. Numerator and denominator use identical months with positive French and global observations. The measure is held fixed thereafter, without clipping, and standardized using the original 55 liable families' mean (16.28%) and SD (26.00 percentage points).

The equal-weighted specification is

$$y_{it} = \alpha_i + \lambda_t + \beta(L_i \times Post_t) + \delta(\widetilde{F}_i \times Post_t) + \gamma(L_i \times \widetilde{F}_i \times Post_t) + \varepsilon_{it}. \tag{1}$$

Every actual liable and control family uses its own standardized share, $\widetilde{F}_i$. The control gradient is δ, the liable gradient is $\delta + \gamma$, and γ measures their difference per original SD. The coefficient β compares the

groups when both have the original liable-sample mean French share. The period, equal family-month weights and parent/control-family resampling blocks follow Equation 10.

For comparability with the business-model heterogeneity regressions, the liable-share-only difference-in-differences specification takes the form

$$y_{it} = \alpha_i + \lambda_t + \beta_D(L_i \times Post_t) + \theta_D(L_i \times Post_t \times \widetilde{F}_i) + \varepsilon_{it}. \tag{2}$$

This benchmark omits the share-by-post term for controls, constraining their exposure gradient to zero. Separate estimates use actual controls and the same preferred synthetic pairs. It is distinct from the own-member-share specification in Equation 1.

For the preferred synthetic specification, the interactions in Equation 1 are estimated using stacked pairs of each liable family and its synthetic comparison, with member fixed effects and shared month fixed effects. The liable member uses its own share; the synthetic member uses the weighted average of its donors' shares, using the same weights as the traffic outcome. For this heterogeneity analysis, the headline French-only traffic-fitting criterion includes $(\widetilde{F}_i - \sum_j w_{ij}\widetilde{F}_j)^2$, a soft French-share-balance penalty with coefficient $\kappa = 1$. Demeaning, donor-change noise scaling and ridge coefficient $\sqrt{12}$ follow the headline fit. Donors require complete French June–November 2018 and January–December 2019 observations, with no German, US or global coverage restriction. Each target can use all 159 residual, 462 exposed or 621 pooled donors in the selected pool. This additional criterion improves French-share balance without imposing equality. Mean absolute share gaps are 2.45, 1.64 and 1.45 percentage points for residual, advertising-exposed and pooled donors, respectively. All 55 liable families remain in the analysis. The French-share penalty is specific to this heterogeneity analysis; headline average effects and joint revenue-share regressions use French-only traffic weights without it.

The monthly version replaces the three post interactions with month-specific interactions. Each bootstrap draw estimates June–November 2018 and January–December 2019 jointly and recenters the coefficients on September–November 2018, retaining the cross-month dependence in the resulting draws. A causal interpretation of γ requires that, absent the tax, the relationship between traffic changes and French share would have been the same in the liable and comparison groups, as well as valid synthetic counterfactuals. Prepolicy patterns and imperfect exposure balance remain relevant diagnostics.

Heterogeneity by French share of global traffic. Figure F.1 separates synthetic controls on the left from equal-weighted actual controls on the right, using unexposed controls in both columns. The upper panels show fitted liable and comparison paths at hypothetical French traffic shares of 0% and 100%, using the original sample mean and standard deviation to standardize the share. The lower panels display the own-member-share differential gradient γ_m and the liable-share-only DiD gradient $\theta_{D,m}$ for each comparison method. Table F.1, Panel A, reports the period coefficients β, δ and γ, together with the liable gradient $\delta+\gamma$. Panel B reports the liable-share-only DiD coefficients β_D and θ_D. The table retains all three control groups

and reports full-pipeline bootstrap standard errors, sample counts and fit statistics for both specifications. French shares use the Amazon.fr calibration conditional on its assumed 90% French traffic share.

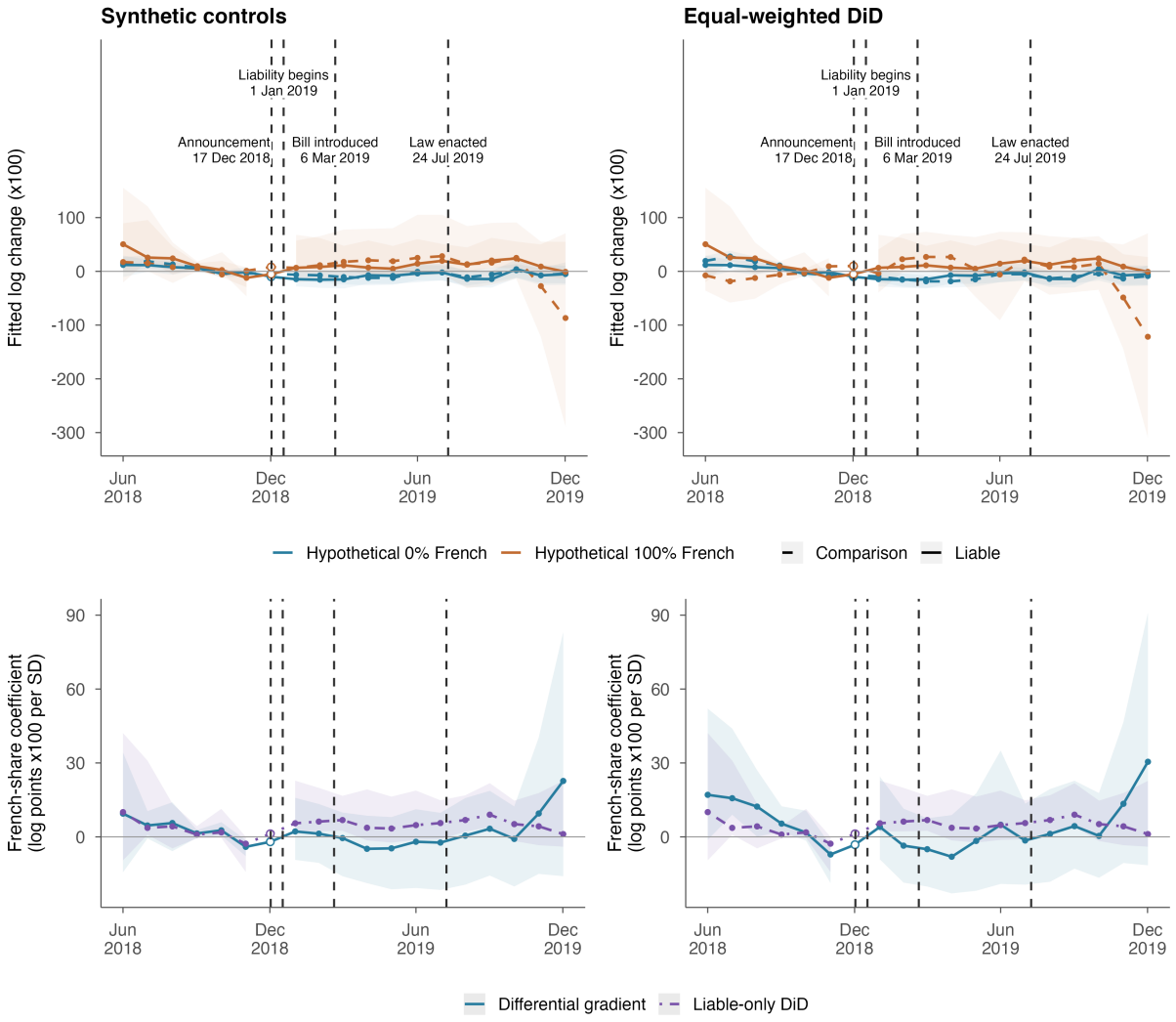


Figure F.1: French-share heterogeneity with traffic-and-exposure-matched synthetic controls

Notes: Left: stacked synthetic controls; right: equal-weighted actual controls, using the residual comparison group. Upper panels show fitted liable and comparison paths at hypothetical 0% and 100% French audience shares, using the original share standardization. Lower panels show the own-member-share differential gradient and the liable-share-only DiD gradient. All paths are centered on September–November 2018 and measured as 100 times log units; gradients are per original SD. Shading gives pointwise 95% percentile intervals from 1,999 full-history parent/family draws. Every draw refits French-only synthetic scaling and weights with the $\kappa = 1$ share-gap penalty, rebuilds outcome-weighted donor shares and baseline offsets, and reestimates both regressions. Donor multiplicities are shared across targets. December is display-only without an interval. Appendix H preserves the conditional figure.

Table F.1: French-share heterogeneity: own-share and DiD specifications

	No observed advertising exposure		Advertising-exposed controls		Pooled controls	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Estimator	Actual	Stacked SC	Actual	Stacked SC	Actual	Stacked SC
Panel A: Own-member-share specification						
Liable × Post (β)	+5.37	−0.90	+8.43	+3.55	+7.56	+3.26
	(7.83)	(7.88)	(7.10)	(7.47)	(6.89)	(7.03)
Share × Post (δ)	+1.96	+3.17	+1.56	+2.19	+1.32	+2.33
	(7.22)	(7.41)	(2.28)	(4.52)	(2.13)	(4.34)
Liable × Share × Post (γ)	+3.25	+2.03	+3.64	+3.01	+3.89	+2.88
	(9.56)	(8.52)	(6.98)	(8.41)	(6.89)	(7.69)
Liable share gradient ($\delta + \gamma$)	+5.20	+5.20	+5.20	+5.20	+5.20	+5.20
	(6.58)	(6.58)	(6.58)	(6.58)	(6.58)	(6.58)
Panel B: Liable-share-only DiD						
Liable × Post (β_D)	+5.74	−0.91	+7.52	+3.46	+7.06	+3.19
	(7.82)	(8.12)	(7.06)	(7.57)	(6.88)	(7.20)
Liable × Share × Post (θ_D)	+5.20	+5.20	+5.20	+5.20	+5.20	+5.20
	(6.58)	(6.58)	(6.58)	(6.58)	(6.58)	(6.58)
Liable families	55	55	55	55	55	55
Liable parent firms	20	20	20	20	20	20
Raw comparison families	169	–	469	–	638	–
Synthetic comparison units	–	55	–	55	–	55
Eligible donor pool	–	159	–	462	–	621
Eligible donors per liable family	–	159	–	462	–	621
Regression units	224	110	524	110	693	110
Observations	3,311	1,650	7,845	1,650	10,331	1,650
Liable / nonliable resampling blocks	20 / 169	20 / 169	20 / 469	20 / 469	20 / 638	20 / 638
Unit fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (A)	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2 (A)	0.00165	0.00818	0.00086	0.00918	0.00063	0.00903
RMSE (log units) (A)	0.369	0.182	0.480	0.178	0.462	0.179
R^2 (B)	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2 (B)	0.00150	0.00647	0.00056	0.00814	0.00043	0.00786
RMSE (log units) (B)	0.369	0.182	0.480	0.178	0.462	0.179

Notes: September–November 2018 versus January–December 2019; December excluded. Coefficients and standard errors are multiplied by 100. Panel A estimates Equation (1); Panel B estimates Equation (2). French shares retain the original 55-family mean (16.28%) and SD (26.00 percentage points), using the fixed Amazon.fr 90% calibration. Synthetic members use outcome-weighted donor shares. In every draw, French-only prepolicy fitting retains the squared standardized share-gap penalty $\kappa = 1$, nonnegative unit-sum weights and ridge coefficient $\sqrt{12}$. All 159, 462 or 621 original eligible donors can enter their resampled pools; no foreign/global coverage restriction is added. The synthetic share regressor is reconstructed from the newly fitted weights. All standard errors are sample standard deviations from 1,999 full-history pairs-bootstrap draws. Each draw samples 20 whole liable parents and samples nonliable families separately within the 469 exposed and 169 residual groups. Family histories and missingness remain together; multiplicities are shared across target and donor roles. All donor-change scales, synthetic weights, baseline offsets and regression coefficients are reestimated. Repeated donor copies receive separate simplex coordinates and ridge penalties. Exposure definitions, original standardization, sample counts and original-sample fit statistics are unchanged. Pointwise intervals use bootstrap percentiles. Appendix H reports the conditional exhibits.

F.2 Joint advertising and intermediation revenue shares

The joint specification includes advertising and intermediation revenue separately, each expressed as a share of total consolidated revenue:

$$\begin{aligned} y_{it} = \alpha_i + \lambda_t + \beta\left(L_i \times Post_t\right) \\ + \theta_A\left(L_i \times Post_t \times \widetilde{A}_{j(i)}\right) + \theta_I\left(L_i \times Post_t \times \widetilde{I}_{j(i)}\right) + \varepsilon_{it}, \end{aligned} \tag{3}$$

where $A_j = R_j^A/R_j$ and $I_j = R_j^I/R_j$, with R_j denoting total consolidated revenue. These shares need not sum to one because firms can earn revenue from other activities. They describe reliance on each revenue source rather than the exact fraction of revenue subject to the French tax.

The supplementary joint revenue-share specification uses the same baseline period, equal family-month weights and fixed effects as the main DiD. Revenue measures are held at fiscal-year 2018 values. Tildes subtract the original liable-family mean and divide by the original liable-family standard deviation. The coefficient β compares groups at mean advertising and intermediation shares; θ_A and θ_I are the respective gradients holding the other share fixed. Share interactions enter only for liable members, including in the stacked synthetic specification.

For Equation 3, consider instead a firm earning all consolidated revenue from advertising, $(A_j, I_j) = (1, 0)$, and a firm earning all consolidated revenue from intermediation, $(A_j, I_j) = (0, 1)$. Their implied responses are

$$\Delta_A^{AI} = \beta + \theta_A \frac{1 - \overline{A}}{s_A} + \theta_I \frac{-\overline{I}}{s_I}, \tag{4}$$

$$\Delta_I^{AI} = \beta + \theta_A \frac{-\overline{A}}{s_A} + \theta_I \frac{1 - \overline{I}}{s_I}. \tag{5}$$

These profiles summarize the fitted relationships at economically interpretable revenue compositions. They are predictions for hypothetical firms and may involve extrapolation beyond the observed combinations of revenue shares. Each full-pipeline bootstrap draw evaluates each profile using the entire coefficient vector. Standard errors are the standard deviation across draws, and confidence intervals are the 2.5th and 97.5th percentiles; this retains dependence among the estimated coefficients.

Advertising and intermediation shares of total revenue. Figure F.2 presents the monthly version of Equation (3), which enters advertising and intermediation shares of total revenue jointly. It compares stacked synthetic controls on the left with equal-weighted actual controls on the right, using controls without observed advertising exposure in both columns. Its upper panels display fitted liable and comparison paths for hypothetical advertising, intermediation and neither-activity firms, defined by $(A, I) = (1, 0)$, $(0, 1)$ and $(0, 0)$. Its lower panels display the corresponding liable-minus-comparison contrasts. Each contrast combines β_m, $\theta_{A,m}$ and $\theta_{I,m}$ evaluated at the stated profile using the original share standardization.

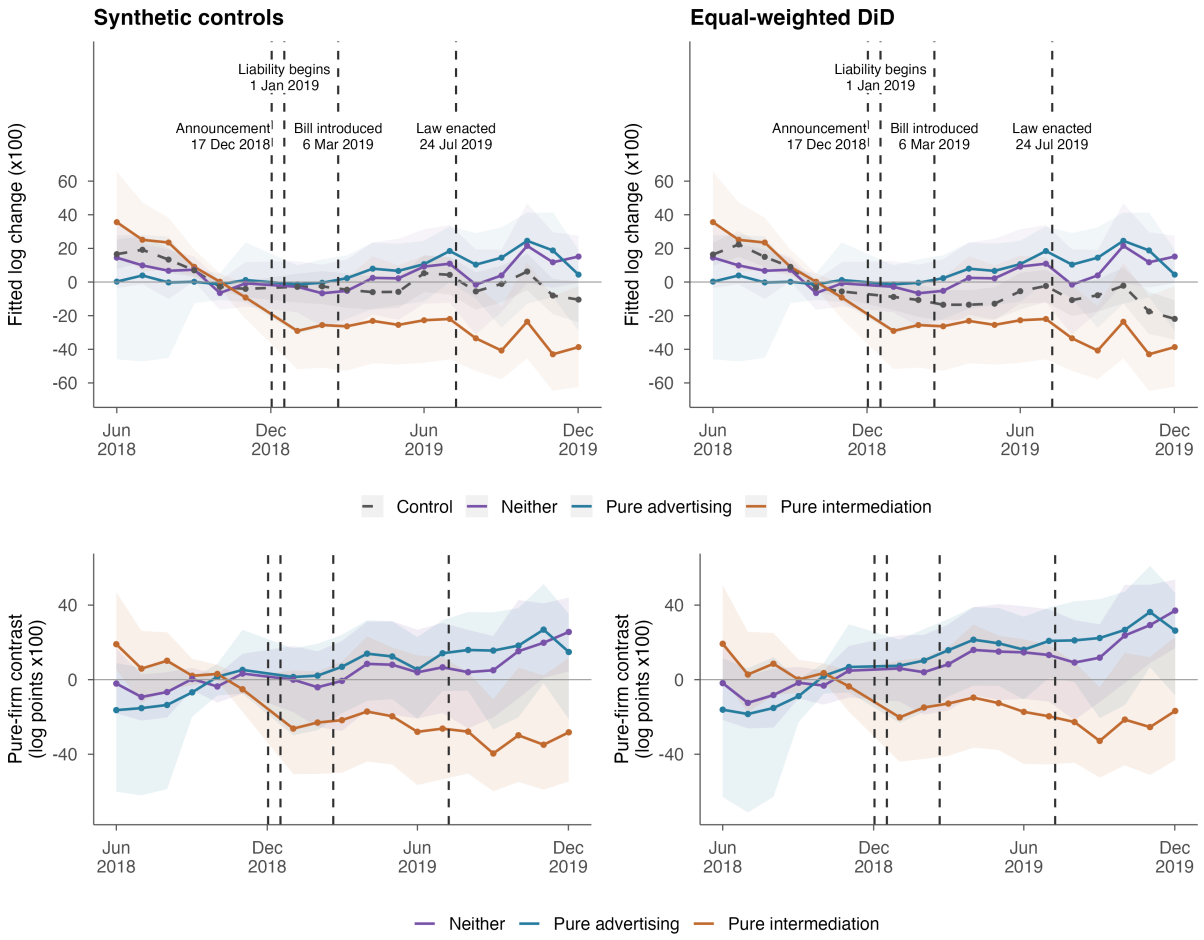


Figure F.2: Advertising and intermediation revenue shares and traffic

Notes: Left: stacked synthetic controls; right: equal-weighted actual controls, using the residual comparison group. Upper panels show fitted paths for hypothetical advertising, intermediation and neither-activity firms, $(A, I) = (1, 0), (0, 1), (0, 0)$, and the comparison path; lower panels show their contrasts. Shares retain original liable-sample standardization. Paths are centered on September–November 2018 in 100 times log units; December is excluded. Shading gives pointwise 95% percentile intervals from 1,999 whole-parent and nonliable-family draws. Every draw refits French-only synthetic scales, traffic-only weights, baseline offsets and the joint revenue-share regression. Original weights do not have a French-share penalty. Shared donor draws retain dependence across targets. Appendix H preserves the conditional figure.

Table F.2 reports β, θ_A and θ_I from Equation (3) for actual-control and stacked synthetic DiD, retaining unexposed, advertising-exposed and pooled controls. Revenue interactions enter only for liable members. Full-pipeline bootstrap standard errors, fit statistics, sample counts and fixed-effect indicators accompany the coefficients. The period comparison is September–November 2018 versus January–December 2019, excluding December 2018. Coefficients are multiplied by 100; share gradients are per original liable-sample SD.

Table F.2: Joint advertising and intermediation revenue-share heterogeneity

	No observed advertising exposure		Advertising-exposed controls		Pooled controls	
Estimator	Actual DiD	Stacked SC	Actual DiD	Stacked SC	Actual DiD	Stacked SC
Model	(1)	(2)	(3)	(4)	(5)	(6)
Liable × Post (β)	+5.74	−2.81	+7.52	+3.75	+7.06	+2.58
	(7.21)	(7.35)	(6.25)	(6.55)	(6.08)	(6.24)
Liable × Post × Advertising (θ_A)	+1.23	+1.23	+1.23	+1.23	+1.23	+1.23
	(5.47)	(5.47)	(5.47)	(5.47)	(5.47)	(5.47)
Liable × Post × Intermediation (θ_I)	−11.69	−11.69	−11.69	−11.69	−11.69	−11.69
	(6.37)	(6.37)	(6.37)	(6.37)	(6.37)	(6.37)
R^2	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2	0.00489	0.03743	0.00141	0.03734	0.00112	0.03662
RMSE (log units)	0.368	0.174	0.480	0.176	0.462	0.175
Liable families / parents	55 / 20	55 / 20	55 / 20	55 / 20	55 / 20	55 / 20
Raw comparison families	169	–	469	–	638	–
Synthetic comparison units	–	55	–	55	–	55
Eligible donor pool	–	159	–	462	–	621
Eligible donors per liable family	–	159	–	462	–	621
Observations	3,311	1,650	7,845	1,650	10,331	1,650
Regression units	224	110	524	110	693	110
Liable / nonliable resampling blocks	20 / 169	20 / 169	20 / 469	20 / 469	20 / 638	20 / 638
Unit fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Equation (3); September–November 2018 versus January–December 2019, excluding December. Coefficients and standard errors are multiplied by 100; gradients are per original liable-sample SD of the fiscal-year 2018 revenue shares. Actual regressions use observed families; stacked comparisons use one liable and one synthetic member per target-month. Synthetic fitting uses French prepolicy traffic, donor-change scaling, nonnegative unit-sum weights and ridge $\sqrt{12}$, without a French-share penalty. The eligible pools contain 159, 462 and 621 French-complete families. All standard errors are sample standard deviations from 1,999 full-history pairs-bootstrap draws. Each draw samples 20 whole liable parents and samples nonliable families separately within the 469 exposed and 169 residual groups. Family histories and missingness remain together; multiplicities are shared across target and donor roles. All donor-change scales, synthetic weights, baseline offsets and regression coefficients are reestimated. Repeated donor copies receive separate simplex coordinates and ridge penalties. Exposure definitions, original standardization, sample counts and original-sample fit statistics are unchanged. Pointwise intervals use bootstrap percentiles. Appendix H reports the conditional exhibits.

G Alternative comparison groups

The comparison groups follow the main-text exposure definition. Exposure requires a positive French-panel detection during September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The residual group includes unknown exposure. The 638 nonliable families divide into 469 exposed and 169 residual families.

Synthetic comparisons use the current French-only traffic fit without French-share balancing. Every target can use all 462 exposed or 621 pooled donors; donor eligibility requires complete French June–November 2018 and January–December 2019 coverage without foreign/global restrictions.

Standard errors follow the main full-pipeline procedure: 1,999 draws resample 20 whole liable parents and nonliable families within exposure groups, preserving histories and missingness. Every draw refits the applicable French-only synthetic weights, donor-change scales, baseline offsets and regressions. Shared donor multiplicities preserve dependence across targets; original composition standardization is retained. Appendix H reports the conditional tables.

G.1 Headline traffic estimates

Table G.1 reports the headline DiD specifications using advertising-exposed and pooled controls. Each control group has an equal-weighted actual-control column and a stacked synthetic-control column. Coefficients, full-pipeline bootstrap standard errors, sample counts and fit statistics use the same specifications and estimation periods as Table 7, which reports the unexposed comparison in the main text.

Table G.1: Headline traffic estimates using advertising-exposed and pooled controls

	Advertising-exposed controls		Pooled controls	
Model	(1)	(2)	(3)	(4)
Estimator	DiD	Stacked SC	DiD	Stacked SC
Dependent variable	y_{it}	y_{ist}	y_{it}	y_{ist}
Liable × Post (β)	+7.52	+3.75	+7.06	+2.58
	(6.82)	(7.11)	(6.62)	(6.76)
Liable families	55	55	55	55
Liable parent firms	20	20	20	20
Raw comparison families	469	–	638	–
Synthetic comparison units	–	55	–	55
Eligible donor pool	–	462	–	621
Eligible donors per liable family	–	462	–	621
Regression units	524	110	693	110
Observations	7,845	1,650	10,331	1,650
Liable / nonliable resampling blocks	20 / 469	20 / 469	20 / 638	20 / 638
Unit fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
R^2	0.911	0.994	0.914	0.994
Within R^2	0.00037	0.00175	0.00027	0.00084
RMSE (log units)	0.480	0.179	0.462	0.178

Notes: January–December 2019 versus September–November 2018; December excluded. Coefficients are 100β. Family/member and month fixed effects, equal family-month weights and the French-only donor eligibility and fitting criterion follow the main specification. Exposed and pooled synthetic donor pools contain 462 and 621 eligible families. All standard errors are sample standard deviations from 1,999 full-history pairs-bootstrap draws. Each draw samples 20 whole liable parents and samples nonliable families separately within the 469 exposed and 169 residual groups. Family histories and missingness remain together; multiplicities are shared across target and donor roles. All donor-change scales, synthetic weights, baseline offsets and regression coefficients are reestimated. Repeated donor copies receive separate simplex coordinates and ridge penalties. Exposure definitions, original standardization, sample counts and original-sample fit statistics are unchanged. Pointwise intervals use bootstrap percentiles. Appendix H reports the conditional exhibits.

G.2 Advertising composition of in-scope revenue

Table G.2 reports β and θ_C from Equation (14), using advertising-exposed and pooled control groups. Actual-control and stacked synthetic DiD columns include full-pipeline bootstrap standard errors, sample counts and fit statistics. Table 9 reports the unexposed comparison in the main text.

Table G.2: Advertising-composition estimates using advertising-exposed and pooled controls

	Advertising-exposed controls		Pooled controls	
Estimator	Actual DiD	Stacked SC	Actual DiD	Stacked SC
Model	(1)	(2)	(3)	(4)
Liable × Post (β)	+7.52	+3.75	+7.06	+2.58
	(5.82)	(6.01)	(5.60)	(5.65)
Liable × Post × Composition (θ_C)	+12.38	+12.38	+12.38	+12.38
	(4.92)	(4.92)	(4.92)	(4.92)
R^2	0.911	0.994	0.914	0.994
Within R^2	0.00147	0.03934	0.00117	0.03864
RMSE (log units)	0.480	0.175	0.462	0.175
Liable families / parents	55 / 20	55 / 20	55 / 20	55 / 20
Raw comparison families	469	–	638	–
Synthetic comparison units	–	55	–	55
Observations	7,845	1,650	10,331	1,650
Regression units	524	110	693	110
Liable / nonliable resampling blocks	20 / 469	20 / 469	20 / 638	20 / 638
Unit fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes

Notes: Equation (14); September–November 2018 versus January–December 2019, excluding December. Coefficients and standard errors are multiplied by 100. Composition retains its original 55-family mean and SD. Actual-control and stacked specifications use equal family-month weights and the headline French-only fitting criterion without French-share balancing. Eligible exposed and pooled donor sets contain 462 and 621 families. All standard errors are sample standard deviations from 1,999 full-history pairs-bootstrap draws. Each draw samples 20 whole liable parents and samples nonliable families separately within the 469 exposed and 169 residual groups. Family histories and missingness remain together; multiplicities are shared across target and donor roles. All donor-change scales, synthetic weights, baseline offsets and regression coefficients are reestimated. Repeated donor copies receive separate simplex coordinates and ridge penalties. Exposure definitions, original standardization, sample counts and original-sample fit statistics are unchanged. Pointwise intervals use bootstrap percentiles. Appendix H reports the conditional exhibits.

H Sensitivity to conditioning on fitted synthetic controls

This appendix reports inference that conditions on fitted synthetic controls. Point estimates, outcomes, donor eligibility, exposure definitions and samples are identical to the main analysis. Only the uncertainty calculation differs. The regression tables and figures use parent-clustered inference for liable families and family-clustered inference for nonliable families, treating the fitted synthetic paths as given. Raw pure-group and empirical Bayes (EB) averages use revenue-stratified whole-parent draws with synthetic weights held fixed: 4,999 draws for annual effects and 1,999 for monthly paths. Each fixed-weight EB draw reestimates model centers, temporal noise and shrinkage parameters.

The main text instead uses 1,999 draws that resample nonliable families as well as liable parents and reconstruct synthetic comparisons in every draw. Ordinary parent draws are used for regressions and model-only profiles; the raw pure-group and EB averages retain the revenue-stratified parent design. Table H.1 reports both the conditional standard errors and fixed-weight bootstrap standard errors calculated on the same draws as the refitted-weight procedure. The paired comparison isolates the effect of donor resampling and weight refitting from the change in inferential method. The variance change need not equal a separate additive first-stage variance, because the paired estimates are dependent.

The composition gradient is identical under fixed and refitted weights within a draw: in the stacked model the synthetic members enter the common comparison intercept, while the composition slope is determined by the liable members.

Table H.1: Annual inference with fixed and refitted synthetic weights

Quantity	Conditional SE	Paired fixed SE	Refitted SE
Liable vs residual	6.38	6.18	7.78
Exposed vs residual	2.98	2.94	4.96
Composition intercept	5.28	5.15	6.99
Composition gradient	4.96	4.92	4.92
Pure advertising / Composition model	7.57	7.75	9.11
Pure advertising / Raw group average	6.60	6.72	7.97
Pure advertising / Empirical Bayes	6.41	6.66	7.81
Pure intermediation / Composition model	9.26	8.94	10.09
Pure intermediation / Raw group average	1.87	1.88	6.59
Pure intermediation / Empirical Bayes	6.35	6.51	8.24

Notes: All standard errors are in log percentage points. Conditional SEs use the clustered or fixed-weight bootstrap methods described above; paired fixed and refitted SEs use identical target draws. Regressions and model endpoints use ordinary draws of 20 whole liable parents, while the raw and EB pure-group averages use revenue-stratified draws of 5 advertising, 10 mixed and 5 intermediation parents. Nonliable families are sampled within the 469 exposed and 169 residual families, sharing multiplicities across target and donor roles. The refitted arm repeats donor eligibility, scaling, fitting, baseline alignment, regressions and EB calculations. The fixed arm retains saved synthetic paths. Both arms use 1,999 draws. Analytical clustered SEs and the 4,999-draw annual EB calculation are not numerically identical to the paired fixed-weight bootstrap. Only five parents belong to each pure group.

Table H.2: Fixed-weight sensitivity: difference-in-differences estimates of French traffic responses

	No documented advertising exposure	
Model	(1)	(2)
Estimator	DiD	Stacked SC
Dependent variable	y_{it}	y_{ist}
Liable × Post (β)	$+5.74$	-2.81
	(7.71)	(6.38)
Liable families	55	55
Liable parent firms	20	20
Raw comparison families	169	–
Synthetic comparison units	–	55
Eligible donor pool	–	159
Eligible donors per liable family	–	159
Regression units	224	110
Observations	3,311	1,650
Clusters	189	20
Unit fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
R^2	0.953	0.994
Within R^2	0.00073	0.00101
RMSE (log units)	0.369	0.177

Notes: January–December 2019 versus September–November 2018; December 2018 excluded. Coefficients are 100β, with clustered standard errors in parentheses. Column (1) estimates Equation (10) on observed families. Column (2) estimates $y_{ist} = \alpha_{is} + \lambda_t + \beta(L_s \times Post_t) + \varepsilon_{ist}$ on one actual liable member and one synthetic member per family-month. Raw columns cluster by liable parent or nonliable family; synthetic pairs share the liable parent cluster. Synthetic inference conditions on fitted weights and omits fitted-weight/shared-donor uncertainty.

Table H.3: Fixed-weight sensitivity: difference-in-differences estimates for advertising-exposed nonliable sites

Model	(1)	(2)
Estimator	Conventional DiD	Stacked synthetic DiD
Dependent variable	y_{it}	y_{ist}
Exposed × Post (δ)	-1.75	-8.24
	(5.52)	(2.98)
Exposed nonliable families	469	469
Raw unexposed families	169	–
Synthetic comparison units	–	469
Starting eligible donor pool	–	159
Eligible donors per exposed family	–	159
Regression units	638	938
Observations	9,506	14,040
Clusters	638	469
Unit fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
R^2	0.895	0.934
Within R^2	0.00004	0.00212
RMSE (log units)	0.477	0.358

Notes: January–December 2019 versus September–November 2018; December 2018 excluded. Coefficients are 100δ (log points times 100); clustered standard errors in parentheses. Column (1) estimates $y_{it} = \alpha_i + \lambda_t + \delta(E_i \times Post_t) + \varepsilon_{it}$ on actual nonliable families, as in Equation (11). Column (2) stacks each exposed family with its own synthetic comparison and estimates $y_{ist} = \alpha_{is} + \lambda_t + \delta(T_s \times Post_t) + u_{ist}$, where $T_s = 1$ for the actual exposed member. Raw standard errors cluster by website family; both members of each synthetic pair share the exposed-family cluster. Synthetic inference conditions on fitted weights and omits fitted-weight/shared-donor uncertainty.

Table H.4: Fixed-weight sensitivity: traffic responses by advertising composition of in-scope revenue

	No documented advertising exposure	
Estimator	Actual DiD	Stacked SC
Model	(1)	(2)
Liable × Post (β)	$+5.74$	-2.81
	(6.82)	(5.28)
Liable × Post × Composition (θ_C)	$+12.38$	$+12.38$
	(4.84)	(4.96)
R^2	0.953	0.994
Within R^2	0.00513	0.03948
RMSE (log units)	0.368	0.173
Liable families / parents	55 / 20	55 / 20
Raw comparison families	169	–
Synthetic comparison units	–	55
Observations	3,311	1,650
Regression units	224	110
Clusters	189	20
Unit fixed effects	Yes	Yes
Month fixed effects	Yes	Yes

Notes: Estimates Equation (14). September–November 2018 versus January–December 2019, excluding December 2018. All coefficients and standard errors are multiplied by 100. Advertising composition uses fiscal-year 2018 accounts and is standardized at the mean and SD across the same 55 liable families. β is the liable-minus-comparison change at mean advertising composition; gradients are per liable-sample SD. Actual regressions cluster by liable parent or control family; synthetic pairs share their liable-parent cluster (20 clusters). Synthetic intervals condition on weights and omit fitted-weight/shared-donor uncertainty.

Table H.5: Fixed-weight sensitivity: pre/post traffic effects by business model and estimation method

	Composition model	Raw group average	Empirical Bayes
Pure advertising	$+16.48$	$+22.25$	$+16.90$
	(7.57)	(6.60)	(6.41)
Pure intermediation	-17.50	$+5.02$	-11.13
	(9.26)	(1.87)	(6.35)

Notes: Entries are annual pre/post traffic effects in log percentage points, with standard errors in parentheses. All effects compare September–November 2018 with January–December 2019; December 2018 is excluded. The composition-model column evaluates the original stacked model at advertising shares $C = 1$ and $C = 0$, using 55 liable website families under 20 parents. The raw column averages website-specific changes in observed-minus-synthetic log traffic among 13 observed pure advertising families and 5 observed pure intermediation families. The EB column averages their empirical Bayes estimates centered on predictions from the original stacked model. Raw and EB averages give equal weight to websites within each group. Model standard errors use the parent-clustered covariance matrix. Raw and EB standard errors use 4,999 revenue-group-stratified whole-parent bootstrap draws; each EB draw reestimates model predictions, noise parameters and residual signal variance. Inference conditions on fitted synthetic weights and omits fitted-weight/shared-donor uncertainty.

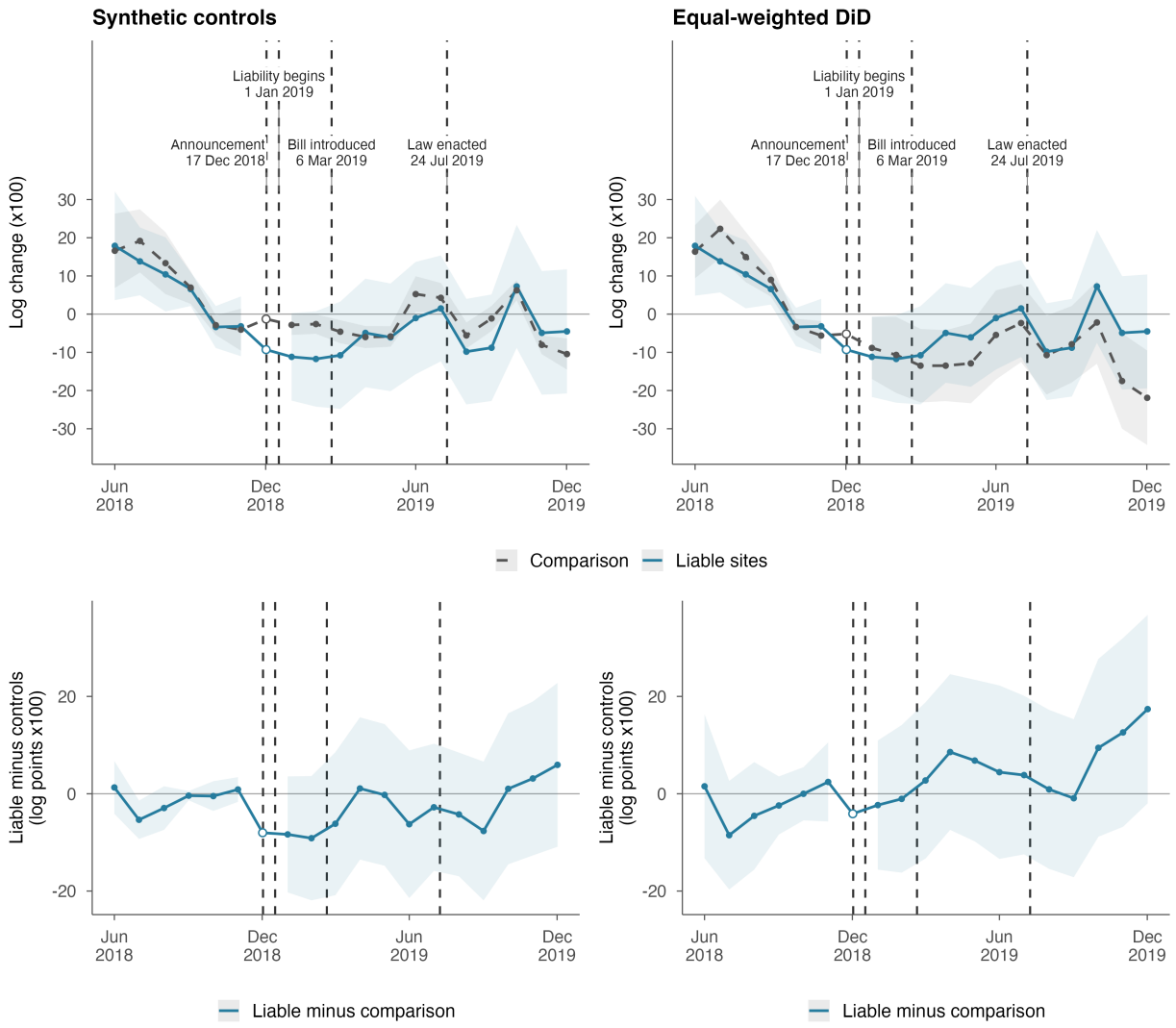


Figure H.1: Fixed-weight sensitivity: french traffic responses of liable platforms

Notes: Left: synthetic controls; right: equal-weighted actual-control DiD. Both compare the same 55 liable families under 20 parents with controls without documented advertising exposure. The raw pool contains 169 families; the starting eligible synthetic donor pool contains 159, with all 159 eligible donors available to every liable family. Upper panels show liable (blue solid) and comparison (gray dashed) paths; lower panels show liable-minus-comparison contrasts. Synthetic weights fit available June–November 2018 French traffic only. All series are centered on September–November 2018 and displayed as 100 times log changes or differences. Scales are common within rows; monthly points lie at calendar midpoints. Shading gives pointwise 95% intervals, clustered by liable parent for synthetic pairs and by liable parent or nonliable family for actual controls; synthetic intervals condition on fitted weights and omit weight/shared-donor uncertainty.

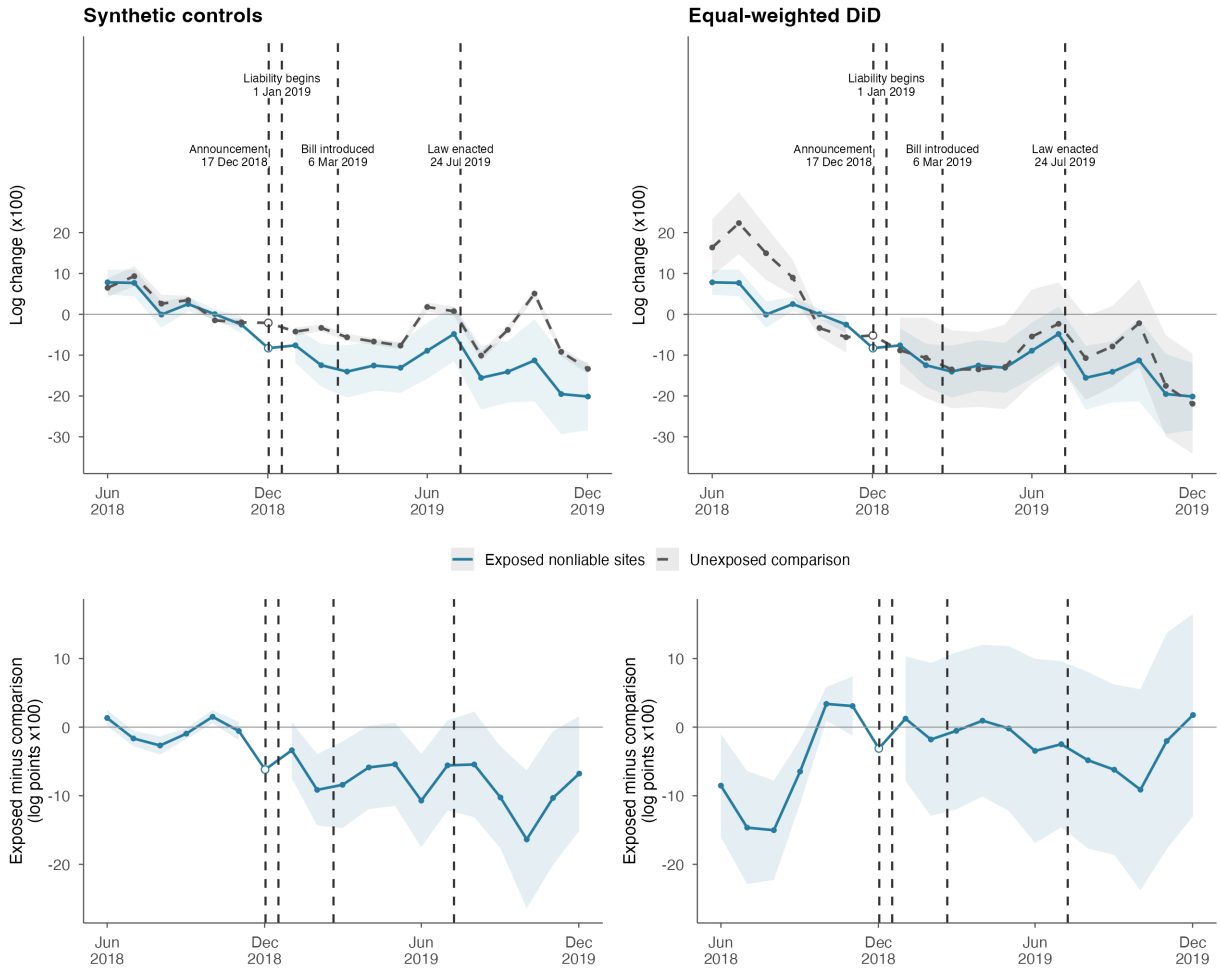


Figure H.2: Fixed-weight sensitivity: french traffic responses of advertising-exposed nonliable sites

Notes: The 469 exposed nonliable website families are compared with 169 families without documented advertising exposure and with individual synthetic comparisons drawn from a starting pool of 159 eligible donors, with all 159 donors eligible for every exposed family. Left column: individual synthetic controls. Right column: equal-weighted actual unexposed controls. Upper row: exposed and comparison paths. Lower row: the corresponding exposed-minus-comparison differences. Monthly outcomes are placed at calendar-month midpoints; connecting lines do not imply daily measurement. Paths are centered on September–November 2018 and displayed as $100b$ for the normalized log change or log difference b. Shading shows pointwise 95% family-clustered intervals, with both members of each synthetic pair assigned to the exposed-family cluster. Synthetic inference conditions on fitted weights and omits fitted-weight/shared-donor uncertainty.



Figure H.3: Fixed-weight sensitivity: advertising composition and traffic

Notes: Left: hypothetical pure advertising ($C = 1$, blue) and pure intermediation ($C = 0$, orange) predictions from the original stacked composition model, estimated on 55 liable families under 20 parents. Middle: raw averages among 13 observed pure advertising families and 5 observed pure intermediation families. Right: empirical Bayes averages for those same groups, centered on predictions from the original model. Available families receive equal weight within each group. Upper panels show normalized traffic paths and synthetic comparisons; the model uses a common comparison path, while the raw and EB columns use each group's own average synthetic path. Lower panels show the corresponding traffic gaps. All paths are centered on September–November 2018 and measured as 100 times log changes or differences. Shading gives pointwise 95% intervals: parent-clustered t intervals for the model and percentile intervals from 1,999 revenue-group-stratified whole-parent draws for the raw and EB columns. Each EB draw reestimates original-model centers, noise parameters and residual signal variances. Inference conditions on fitted synthetic weights and omits fitted-weight/shared-donor uncertainty.

H.1 Conditional inference for the supplementary specifications

The following exhibits report fixed-weight supplementary analyses: the own-member French-share and liable-share-only specifications, joint revenue shares, and alternative comparison groups. Their coefficients and fitted paths are identical to the corresponding exhibits in Appendices F and G; their standard errors and pointwise intervals condition on saved weights. The refitted-weight French-share analysis additionally reestimates the $\kappa = 1$ share-penalized weights and the outcome-weighted donor-share regressor in each bootstrap draw. Joint revenue-share and alternative-control analyses refit the headline traffic-only weights. The conditional exhibits below use parent/family-clustered inference as sensitivity checks.

Table H.6: Fixed-weight sensitivity: French-share heterogeneity: own-share and DiD specifications

	No observed advertising exposure		Advertising-exposed controls		Pooled controls	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Estimator	Actual	Stacked SC	Actual	Stacked SC	Actual	Stacked SC
Panel A: Own-member-share specification						
Liable × Post (β)	+5.37	−0.90	+8.43	+3.55	+7.56	+3.26
	(7.79)	(6.14)	(7.00)	(6.59)	(6.78)	(6.35)
Share × Post (δ)	+1.96	+3.17	+1.56	+2.19	+1.32	+2.33
	(6.64)	(0.92)	(2.21)	(1.95)	(2.07)	(2.00)
Liable × Share × Post (γ)	+3.25	+2.03	+3.64	+3.01	+3.89	+2.88
	(7.00)	(2.26)	(3.13)	(3.19)	(3.03)	(3.13)
Liable share gradient ($\delta + \gamma$)	+5.20	+5.20	+5.20	+5.20	+5.20	+5.20
	(2.22)	(2.28)	(2.21)	(2.28)	(2.21)	(2.28)
Panel B: Liable-share-only DiD						
Liable × Post (β_D)	+5.74	−0.91	+7.52	+3.46	+7.06	+3.19
	(7.82)	(6.31)	(6.95)	(6.70)	(6.76)	(6.47)
Liable × Share × Post (θ_D)	+5.20	+5.20	+5.20	+5.20	+5.20	+5.20
	(2.22)	(2.28)	(2.21)	(2.28)	(2.21)	(2.28)
Liable families	55	55	55	55	55	55
Liable parent firms	20	20	20	20	20	20
Raw comparison families	169	–	469	–	638	–
Synthetic comparison units	–	55	–	55	–	55
Eligible donor pool	–	159	–	462	–	621
Eligible donors per liable family	–	159	–	462	–	621
Regression units	224	110	524	110	693	110
Observations	3,311	1,650	7,845	1,650	10,331	1,650
Clusters	189	20	489	20	658	20
Unit fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (A)	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2 (A)	0.00165	0.00818	0.00086	0.00918	0.00063	0.00903
RMSE (log units) (A)	0.369	0.182	0.480	0.178	0.462	0.179
R^2 (B)	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2 (B)	0.00150	0.00647	0.00056	0.00814	0.00043	0.00786
RMSE (log units) (B)	0.369	0.182	0.480	0.178	0.462	0.179

Notes: September–November 2018 versus January–December 2019; December 2018 excluded. Coefficients and clustered standard errors are multiplied by 100; share gradients are per original liable-sample SD (26.00 percentage points). Panel A estimates Equation (1), and Panel B estimates Equation (2). Shares use matched pre-announcement French/global observations and the Amazon.fr calibration conditional on 90% French traffic, standardized at the original liable mean (16.28%) and SD. Actual families use their own shares; synthetic members use outcome-weighted donor shares. Synthetic weights fit available June–November 2018 French traffic, with demeaning, donor-change noise scaling, ridge coefficient $\sqrt{12}$ and a squared standardized French-share-gap penalty ($\kappa = 1$). Donors require complete French June–November 2018 and January–December 2019 observations; there is no foreign/global coverage restriction. All targets have access to every donor in the selected pool; share balance remains approximate. Actual comparisons cluster by liable parent/control family; synthetic pairs share the liable-parent cluster. Synthetic inference conditions on fitted weights and omits fitted-weight/shared-donor uncertainty. Exposure requires a positive French-panel September–November 2018 detection on any family member domain of a study-designated liable provider in the WTM advertising category, excluding adult advertising and custom overrides. The residual group includes unknown exposure. Fit statistics do not establish counterfactual validity.

Table H.7: Fixed-weight sensitivity: Joint advertising and intermediation revenue-share heterogeneity

	No observed advertising exposure		Advertising-exposed controls		Pooled controls	
Estimator	Actual DiD	Stacked SC	Actual DiD	Stacked SC	Actual DiD	Stacked SC
Model	(1)	(2)	(3)	(4)	(5)	(6)
Liable × Post (β)	+5.74	−2.81	+7.52	+3.75	+7.06	+2.58
	(6.59)	(4.72)	(5.53)	(4.81)	(5.29)	(4.68)
Liable × Post × Advertising (θ_A)	+1.23	+1.23	+1.23	+1.23	+1.23	+1.23
	(2.92)	(3.00)	(2.91)	(3.00)	(2.91)	(3.00)
Liable × Post × Intermediation (θ_I)	−11.69	−11.69	−11.69	−11.69	−11.69	−11.69
	(5.75)	(5.90)	(5.73)	(5.90)	(5.73)	(5.90)
R^2	0.953	0.994	0.911	0.994	0.914	0.994
Within R^2	0.00489	0.03743	0.00141	0.03734	0.00112	0.03662
RMSE (log units)	0.368	0.174	0.480	0.176	0.462	0.175
Liable families / parents	55 / 20	55 / 20	55 / 20	55 / 20	55 / 20	55 / 20
Raw comparison families	169	–	469	–	638	–
Synthetic comparison units	–	55	–	55	–	55
Eligible donor pool	–	159	–	462	–	621
Eligible donors per liable family	–	159	–	462	–	621
Observations	3,311	1,650	7,845	1,650	10,331	1,650
Regression units	224	110	524	110	693	110
Clusters	189	20	489	20	658	20
Unit fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimates Equation (3). September–November 2018 versus January–December 2019, excluding December 2018. All coefficients and standard errors are multiplied by 100. Revenue measures use fiscal-year 2018 accounts and are standardized at the mean and SD across the same 55 liable families. β is the liable-minus-comparison change at mean revenue shares; gradients are per liable-sample SD. All website-months receive equal weight. Actual DiD uses observed nonliable families; stacked SC includes one liable and one synthetic member per target-month. Nonnegative weights summing to one fit available June–November 2018 French histories with demeaning, donor-change noise scaling and ridge coefficient $\sqrt{12}$; there is no French-share penalty. Donors require complete French June–November 2018 and January–December 2019 observations, without foreign or global coverage requirements. Every target can use all 159, 462 or 621 donors in its selected pool. Actual regressions cluster by liable parent or control family; synthetic pairs share their liable-parent cluster (20 clusters). Synthetic intervals condition on weights and omit fitted-weight/shared-donor uncertainty. Overall and within R^2 and RMSE are reported separately for each specification. Advertising exposure requires a positive French-panel detection in September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The 638 nonliable families comprise 469 exposed and 169 with no observed qualifying exposure, including unknown exposure.

Table H.8: Fixed-weight sensitivity: Headline traffic estimates using advertising-exposed and pooled controls

	Advertising-exposed controls		Pooled controls	
Model	(1)	(2)	(3)	(4)
Estimator	DiD	Stacked SC	DiD	Stacked SC
Dependent variable	y_{it}	y_{ist}	y_{it}	y_{ist}
Liable × Post (β)	+7.52	+3.75	+7.06	+2.58
	(6.83)	(6.29)	(6.63)	(6.21)
Liable families	55	55	55	55
Liable parent firms	20	20	20	20
Raw comparison families	469	–	638	–
Synthetic comparison units	–	55	–	55
Eligible donor pool	–	462	–	621
Eligible donors per liable family	–	462	–	621
Regression units	524	110	693	110
Observations	7,845	1,650	10,331	1,650
Clusters	489	20	658	20
Unit fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
R^2	0.911	0.994	0.914	0.994
Within R^2	0.00037	0.00175	0.00027	0.00084
RMSE (log units)	0.480	0.179	0.462	0.178

Notes: January–December 2019 versus September–November 2018; December 2018 excluded. Coefficients are 100β, with clustered standard errors in parentheses. Actual comparisons use family and month fixed effects; stacked synthetic comparisons use one actual liable and one synthetic member per family-month, with member and shared month fixed effects. Synthetic weights fit available June–November 2018 French histories with demeaning, donor-change noise scaling, nonnegative unit-sum weights and ridge coefficient $\sqrt{12}$. Donors require complete French June–November 2018 and January–December 2019 observations, without foreign/global coverage restrictions; every target can use all 462 exposed or 621 pooled donors. The headline fitting procedure and validation are described in Section 4. Raw columns cluster by liable parent or nonliable family; synthetic pairs share the liable-parent cluster. Inference conditions on fitted weights and omits fitted-weight/shared-donor uncertainty. Exposure requires a positive French-panel September–November 2018 detection on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded; unknown exposure remains in the residual group. Overall R^2 includes fixed effects; within R^2 absorbs them. Fit statistics do not establish counterfactual validity.

Table H.9: Fixed-weight sensitivity: Advertising-composition estimates using advertising-exposed and pooled controls

	Advertising-exposed controls		Pooled controls	
Estimator	Actual DiD	Stacked SC	Actual DiD	Stacked SC
Model	(1)	(2)	(3)	(4)
Liable × Post (β)	+7.52	+3.75	+7.06	+2.58
	(5.81)	(4.78)	(5.57)	(4.77)
Liable × Post × Composition (θ_C)	+12.38	+12.38	+12.38	+12.38
	(4.82)	(4.96)	(4.82)	(4.96)
R^2	0.911	0.994	0.914	0.994
Within R^2	0.00147	0.03934	0.00117	0.03864
RMSE (log units)	0.480	0.175	0.462	0.175
Liable families / parents	55 / 20	55 / 20	55 / 20	55 / 20
Raw comparison families	469	–	638	–
Synthetic comparison units	–	55	–	55
Observations	7,845	1,650	10,331	1,650
Regression units	524	110	693	110
Clusters	489	20	658	20
Unit fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes

Notes: Estimates Equation (14). September–November 2018 versus January–December 2019, excluding December 2018. All coefficients and standard errors are multiplied by 100. Revenue measures use fiscal-year 2018 accounts and are standardized at the mean and SD across the same 55 liable families. β is the liable-minus-comparison change at mean revenue shares; gradients are per liable-sample SD. All website-months receive equal weight. Actual DiD uses observed nonliable families; stacked SC includes one liable and one synthetic member per target-month. Nonnegative weights summing to one fit available June–November 2018 French histories with demeaning, donor-change noise scaling and ridge coefficient $\sqrt{12}$; there is no French-share penalty. Donors require complete French June–November 2018 and January–December 2019 observations, without foreign or global coverage requirements. Every target can use all 462 exposed or 621 pooled donors in its selected pool. Actual regressions cluster by liable parent or control family; synthetic pairs share their liable-parent cluster (20 clusters). Synthetic intervals condition on weights and omit fitted-weight/shared-donor uncertainty. Overall and within R^2 and RMSE are reported separately for each specification. Advertising exposure requires a positive French-panel detection in September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The 638 nonliable families comprise 469 exposed and 169 with no observed qualifying exposure, including unknown exposure.

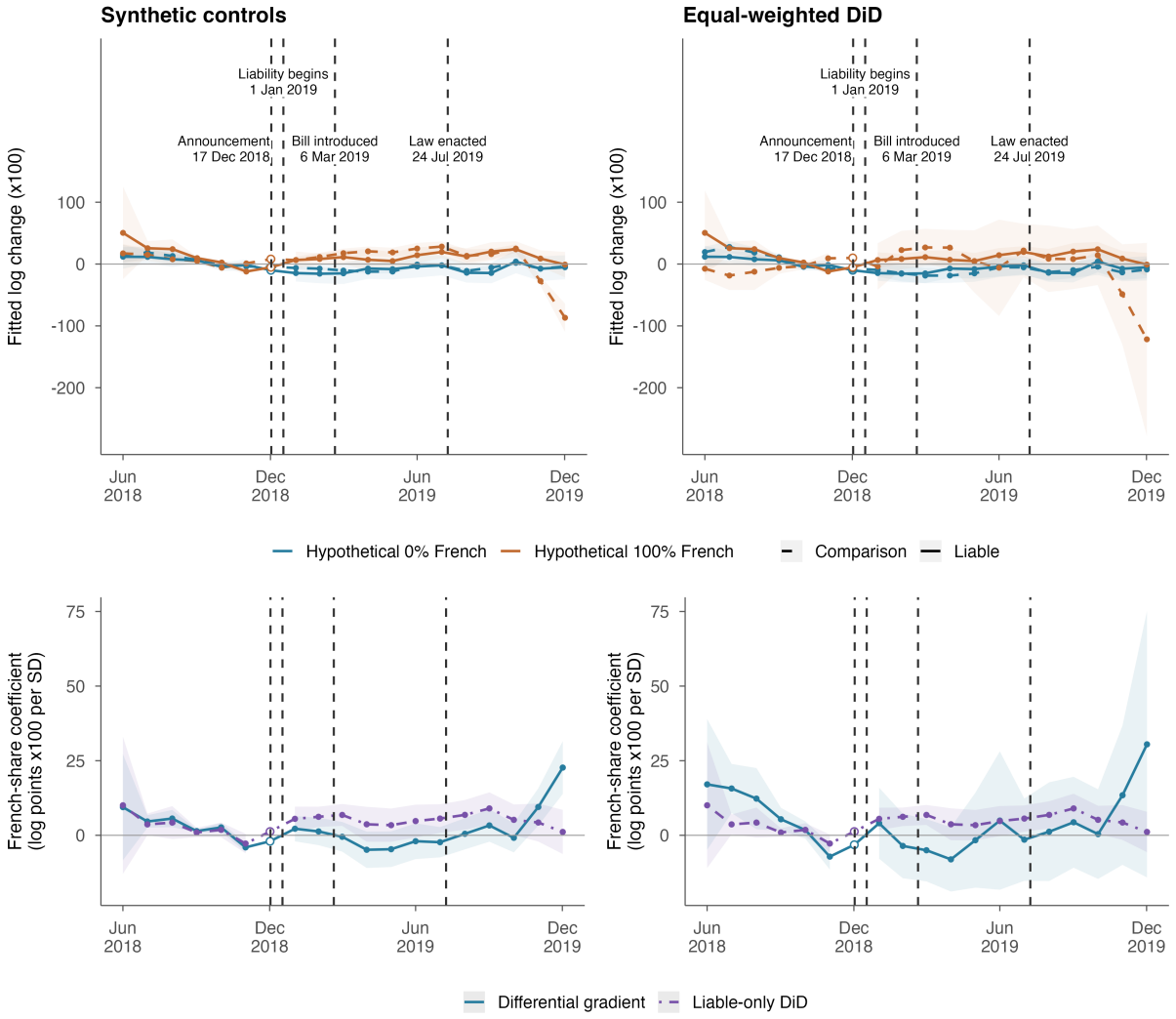


Figure H.4: Fixed-weight sensitivity: French-share heterogeneity with traffic-and-exposure-matched synthetic controls

Notes: Left: synthetic controls; right: equal-weighted actual-control DiD. Both use controls without documented advertising exposure and retain 55 liable families under 20 parents. Upper panels show fitted liable (solid) and comparison (dashed) paths at hypothetical 0% and 100% French traffic shares, using each column's own-member-share model and the original standardization. These endpoint profiles may extrapolate beyond observed support; they do not change the estimation sample. Lower panels show the differential gradient γ_m (blue; Table F.1, Panel A) and liable-share-only DiD gradient $\theta_{D,m}$ (purple; Panel B), omitting control-share interactions. Each purple curve uses its column's comparison method. Table F.1 retains advertising-exposed and pooled comparisons. Synthetic weights fit available June–November 2018 French traffic plus a share-gap penalty ($\kappa = 1$), with demeaning, donor-change noise scaling and ridge coefficient $\sqrt{12}$; balance is approximate. All 159 residual donors are eligible for every target, requiring complete French pre/post coverage without foreign/global restrictions. Shares use the Amazon.fr calibration conditional on 90% French traffic. Values are 100 times log coefficients; gradients are per original share SD (26.00 percentage points). Models jointly estimate June–November 2018 and January–December 2019, centered on September–November using full covariance. Hollow December points are auxiliary and display-only. Shading gives 95% clustered t intervals; synthetic intervals condition on weights and omit weight/shared-donor uncertainty. Monthly points lie at calendar midpoints; labeled lines mark announcement, liability, bill introduction and enactment. Exposure requires a positive French-panel detection during September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The residual group includes unknown exposure.

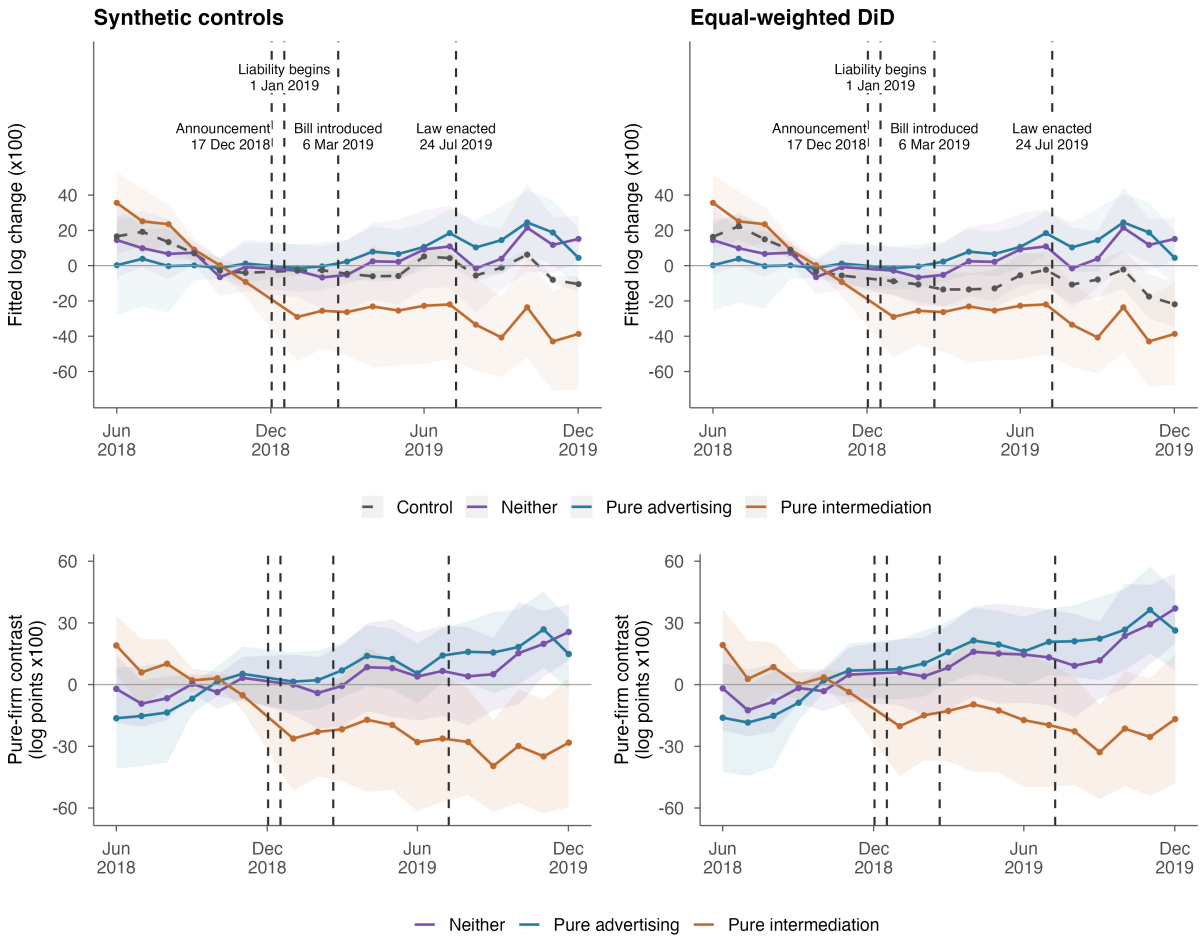


Figure H.5: Fixed-weight sensitivity: Advertising and intermediation revenue shares and traffic

Notes: Both columns use controls without observed advertising exposure and retain 55 liable families under 20 parents. Left: stacked synthetic-control DiD; right: equal-weighted actual-control DiD, corresponding to columns (2) and (1) of Table F.2. Upper panels show fitted comparison time effects (gray dashed) and hypothetical liable paths at $(A, I) = (1, 0)$ (pure advertising, blue), $(0, 1)$ (pure intermediation, orange), and $(0, 0)$ (neither activity, purple). These hypothetical profiles may extrapolate beyond observed shares; the neither profile is evaluated under liability despite zero in-scope revenue shares. Lower panels show liable-minus-control contrasts $\Delta_{A,m} = \beta_m + \theta_{A,m}(1 - \bar{A})/s_A - \theta_{I,m}\bar{I}/s_I$ and $\Delta_{I,m} = \beta_m - \theta_{A,m}\bar{A}/s_A + \theta_{I,m}(1 - \bar{I})/s_I$, and $\Delta_{N,m} = \beta_m - \theta_{A,m}\bar{A}/s_A - \theta_{I,m}\bar{I}/s_I$. All values are 100 times log coefficients, centered on September–November 2018 using full covariance. Models jointly estimate June–November 2018 and January–December 2019; December is excluded. Shading gives 95% clustered t intervals; synthetic intervals condition on fitted weights and omit weight/shared-donor uncertainty. Monthly points lie at calendar midpoints. Event lines mark 17 December 2018 (announcement), 1 January 2019 (liability), 6 March 2019 (bill) and 24 July 2019 (enactment). Synthetic weights use the headline French-only traffic criterion without French-share matching, with demeaning, donor-change noise scaling and ridge coefficient $\sqrt{12}$. All 159 residual donors are eligible for every target under complete French pre/post coverage without foreign/global restrictions. Exposure requires a positive French-panel detection during September–November 2018 on any family member domain of a study-designated liable provider in the WTM advertising category. Adult advertising and custom overrides are excluded. The residual group includes unknown exposure.

References

Adevinta (2023). Annual report 2022. https://adevinta.com/wp-content/uploads/2023/04/Adevinta-AR2022.pdf. Note 21.3.1, p. 147.

ANGI Homeservices (2020). Form 10-k for the year ended december 31, 2019. https://www.sec.gov/Archives/edgar/data/1705110/000170511020000003/angi-20193112x10k.htm. Management discussion and analysis.

Assemblée Nationale (2023). Rendement de la taxe sur les services numériques. Journal Officiel. Written question no. 2974 (C. Sitzenstuhl); response from the Ministère de l'Économie published in the Journal Officiel, 14 February 2023, p. 1448. Firm counts sourced to the DGFiP.

Booking Holdings (2020). Form 10-k for the year ended december 31, 2019. https://www.sec.gov/Archives/edgar/data/1075531/000107553120000011/bkng1231201910k.htm. Management discussion and analysis.

Bureau van Dijk, a Moody's Analytics company (2025). Orbis. Global company database covering financials, ownership structures, and subsidiaries.

IAC/InterActiveCorp (2020). Form 10-k for the year ended december 31, 2019. https://www.sec.gov/Archives/edgar/data/891103/000089110320000004/iac-20191231x10k.htm. Management discussion and analysis.

S&P Global Market Intelligence (2025). Compustat. Standardized financial statements for publicly traded firms; accessed via Wharton Research Data Services (WRDS).

TripAdvisor (2020). Form 10-k for the year ended december 31, 2019. https://www.sec.gov/Archives/edgar/data/1526520/000156459020005269/trip-10k_20191231.htm. Management discussion and analysis.

United States Trade Representative (2019). Report on france's digital services tax. Technical report, Office of the United States Trade Representative.